COURSE BOOK B.TECH. II YEAR

(Autonomous)

Computer Science

www.kiet.edu
Delhi-NCR, Ghaziabad

KIET
GROUP OF INSTITUTIONS
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CELEBRATING 25 YEARS OF EXCELLENCE

CURRICULUM STRUCTURE & SYLLABUS

Effective from the Session: 2026-27

VISION

To become a leading institution nationally in the area of professional education, research & innovation for serving the global community.

MISSION

- To impart quality professional education, skills and values through outcome-based innovative teaching learning process in all spheres.
- To undertake collaborative interdisciplinary research as a co-requisite for professional education and simultaneously solve problems faced by society and industry.
- To create an ambience of innovation, entrepreneurship and consultancy for future leaders and innovators.
- To keep faculty members enthusiastic by continuous professional development and positive working environment.

Department of Computer Science

VISION

To emerge as a leader in the field of computer science education with innovation and research to create a positive global impact.

MISSION

- To provide quality education in computer science to shape next-generation leaders for global community.
- To equip students with skills and knowledge to bridge the gap between academia and industry by integrating the latest technologies and innovative practices.
- To create a conducive environment for research and innovation for providing sustainable solutions.
- To develop socially responsible professionals while encouraging personal and professional growth.

PEOs

Graduates will be able to-

- **PEO-1 :** Engaged in successful career in the software industry and higher Studies.
- **PEO-2 :** Adaptable to the recent trends for developing innovative solutions.
- **PEO-3 :** Socially responsible global citizen and leaders in their domain.

B.Tech. Computer Science

Program Outcomes (Pos)

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability fori) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

- **PSO1:** Problem-Solving Skills: Provide effective and efficient real time solutions using acquired knowledge with the help of algorithms, architecture, and other modern technologies.
- **PSO2:** Professional Skills: Apply the relevant contemporary trends in industrial and research domain through extended practices.

B.Tech (CS) 3rd Sem

S No.	Course Category (AICTE)	Course Category (UGC)	BOS	Course Code	Course Name	Type	Academic Learning (AL)			Continuous Internal Examination (CIE)			End Sem Examination (ESE)	Total Marks	Total Credits
							L	T	P	MSE	CA	TOTAL		CIE+ ESE	
1	PC	Major (Core)	IT	IT301L	Database Systems	L	3	0	0	60	15	75	75	150	3
2	PC	Major (Core)	CSE	CS206L	Operating System	L	3	0	0	60	15	75	75	150	3
3	BS	Major (Core)	ASH	MA105L	Probability and Statistics	L	3	0	0	60	15	75	75	150	3
4	MC	Value Added	ASH	HS109L	Constitution of India	L	2	0	0	25	-	25	25	-	NC
5	HS	AEC	ASH	HS110L	Aptitude-1	L	1	0	0	-	25	25	25	50	1
6	HS	AEC	ASH	HS111L	Soft Skills Essential-1	L	1	0	0	-	25	25	25	-	NC
Blended															
7	PC	Major (Core)	CS	CS336B	Object Oriented Programming with Java	B	3	0	2	80	20	100	100	200	4
8	PC	Major (Core)	CSE	CS302B	Advance Data Structure	B	3	0	2	80	20	100	100	200	4
9	PC	Major (Core)	CS	CS205B	Artificial Intelligence and It's Applications	B	3	0	2	80	20	100	100	200	4
Lab/Practical															
10	PC	Major (Core)	IT	IT301P	Database Systems Lab	P	0	0	2	-	25	25	25	50	1
11	PC	Major (Core)	CSE	CS206P	Operating System Lab	P	0	0	2	-	25	25	25	50	1
12	PW	Summer Internship	CSIT	IT105P	Social Internship Assessment	P	0	0	0	-	50	50	-	50	1
Total Hours : 32 hrs.							**22**	**0**	**10**					**1250**	**25**

B.Tech (CS) 4th Sem

S No.	Course Category (AICTE)	Course Category (UGC)	BOS	Course Code	Course Name	Type	Academic Learning (AL)			Continuous Internal Examination (CIE)			End Sem Examination (ESE)	Total Marks	Total Credits
							L	T	P	MSE	CA	TOTAL		CIE+ ESE	
1	PC	Major (Core)	CSE	CS401L	Design and Analysis of Algorithms	L	3	0	0	60	15	75	75	150	3
2	ES	Major (Core)	IT	IT302L	Computer Networks	L	3	0	0	60	15	75	75	150	3
3	MC	Value Added	ASH	HS112L	Universal Human Values	L	3	0	0	60	15	75	75	150	3
4	HS	AEC	ASH	HS113L	Aptitude-2	L	1	0	0	-	25	25	25	50	1
5	HS	AEC	ASH	HS114L	Soft Skills Essential-2	L	1	0	0	-	25	25	25	-	NC
Blended															
6	PC	Major (Core)	CSIT	IT202B	Data Analytics	B	2	0	2	60	15	75	75	150	3
7	PC	Major (Core)	CS	CS303B	ANN and Machine Learning	B	2	0	2	60	15	75	75	150	3
8	PC	Major (Core)	CSE	CS208B	Web Technology	B	2	0	2	60	15	75	75	150	3
9	PE	Major (Core)/ SEC	-	-	Professional Elective-I	B	3	0	2	80	20	100	100	200	4
Lab/Practical															
10	PC	Major (Core)	CSE	CS401P	Design and Analysis of Algorithms Lab	P	0	0	2	-	25	25	25	50	1
11	ES	Major (Core)	IT	IT302P	Computer Networks Lab	P	0	0	2	-	25	25	25	50	1
Total Hours : 32 hrs.							**20**	**0**	**12**					**1250**	**25**

Professional Electives (PE)

Course Type (PE)	**Full Stack Product Engineering with AI** *(Powered by Xebia)*	**Next-Gen AI: From Models to Agents** *(Powered by Xebia)*	**DevOps and Autonomous Engineering** *(Powered by Xebia)*	**Data Engineering** (Powered by AWS)	**Applied Apple Technologies: Swift, XCode & iOS Development** *(Powered by Apple)*	**Cyber Security** (Powered by NSDC)
BOS	**CSE**	**CSE**	**CSE**	**CSE**	**CSE**	**CSIT**
PE I- (4th Sem)	Frontend Engineering with React & Next (CS318E)	Intelligent Systems with Text & Vision API (CS307E)	DevOps Foundations & Version Control (CS304E)	AWS Foundations & Data Engineering (CS335E)	Foundation of iOS App Development (CS321E)	Microsoft Azure Fundamentals (IT306E)

Theory Courses Detail Syllabus – III Sem

Course Code: IT301L	Course Name: Database Systems	L	T	P	C
		3	0	0	3

Pre-requisite: Concepts of any programming language

Course Objectives:
1. To develop a strong foundation in database management concepts.
2. To equip students with practical skills in database design, normalization, transaction management, and recovery techniques.

Course Outcome: After completion of the course, the student will be able to
1. Acquire knowledge of database design methodology for real-life applications
2. Design an information model using the concept of ER diagram
3. Apply the concept of SQL on real-life databases
4. Analyze the redundancy problem in the database and reduce it using normalization.
5. Identify the broad range of database management issues including data integrity, security, and recovery transactions, as well as enforce entity integrity, referential integrity, key constraints, and domain constraints on the database.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	1	2	1	1	1	1	1	2
CO2	2	2	3	2	3	1	1	2	2	2	2
CO3	3	3	2	1	3	1	1	1	1	1	2
CO4	3	3	2	2	2	1	1	1	1	1	2
CO5	3	3	2	2	3	2	2	1	1	1	2

Unit 1 | Introduction to Database System | 09 hours

Database System vs File System, Database System Concept and Architecture, Data Model Schema and Instances, Data Independence and its Types, Overall Database Structure.

Entity Relationship Model: ER Model Concepts, Notation for ER Diagram, Mapping Constraints, Key attribute, Generalization, Aggregation, Reduction of an ER Diagrams to Tables.

Use Case: Discuss one or two case studies like Banking System, and National Hockey League (NHL).

Unit 2 | Relational data Model | 09 hours

Relational Data Model Concepts, type of keys, Integrity Constraints- Entity integrity and referential integrity, Keys Constraints, Domain Constraints, Relational Algebra-Unary Relational Operations- SELECT and PROJECT, Binary Relational Operations-CROSS, JOIN and DIVISION, Queries in Relational Algebra.

Database Implementation using SQL: Introduction to SQL, Characteristics of SQL, SQL Data Types, Basic Queries in SQL- create, select Insert, Delete and Update Statements, concepts of group by and having, order by, Sub Queries, Aggregate Functions, Joins, Unions, Intersection, Minus, Views.

Note: Hands-on in the class/ Lab

Unit 3 | Database Design and Normalization | 09 hours

Functional Dependencies, Inference rules, Closure of attributes, FD equivalence and Minimal cover.

Normalization: Normal forms, first, second, third normal forms, and BCNF. Lossless join decompositions, Dependency Preservation, Multi-valued Dependencies and Fourth Normal Form, Join Dependencies and Fifth Normal Form.

Note: Discuss one or two case studies for the Normalization. e.g.
1. In this a table would be given and students will be asked to normalize in a higher normal form
2. Schema along with FDs set will be given and students will be asked to normalize in a higher normal form

Unit 4 | Transaction Processing | 09 hours

Transaction and its States, ACID property, Transaction Scheduling, Serializability of scheduling, Conflict, and View Serializability

Concurrency Control Techniques: Concurrency Control, Locking Techniques for Concurrency Control, Two-phase locking techniques for concurrency control, Time Stamping Protocols for Concurrency Control

Unit 5 | Database Recovery Techniques | 09 hours

Recovery Concepts, Recoverability, Log Based Recovery, Checkpoints, Shadow Paging, The ARIES recovery, Deadlock Handling

PL/SQL: Introduction, features, syntax, DDL within Pl/SQL, DML in PL/SQL, Cursors, stored procedures, stored function, database triggers, indexing, **Case Study-Microsoft Azure SQL.**

Note: Hands-on in the class/ Lab

Total Lecture Hours | 45 hours

Textbook:
1. Elmasri, Navathe, " Fundamentals of Database Systems", Addision Wesley
2. Korth, Silbertz, Sudarshan," Database Concepts", McGraw Hill
3. Date C J, "An Introduction to Database Systems", Addision Wesley

Reference Books:

1. Bipin C. Desai, "An Introduction to Database Systems", Gagotia Publications 8. Majumdar & Bhattacharya, "Database Management System", TMH
2. RAMAKRISHNAN"Database Management Systems", McGraw Hill
3. Rafe Colburn, Using SQL, PHI

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
60	15	75	150
75			

Course Code: CS206L	Course Name: Operating System	L	T	P	C
		3	0	0	3

Pre-requisite: Basic knowledge on Computer System and system memory, Computer Organization and Logic Design (COLD)

Course Objectives:
This course aims to provide a comprehensive understanding of operating systems, their components, structures, and functionalities. It covers fundamental concepts such as process management, memory management, file systems, and concurrency. The course also emphasizes practical aspects, such as system calls, Linux commands, shell scripting, and scheduling algorithms. By the end of this course, students will have the knowledge and skills to understand, analyze, and apply operating system principles in real-world scenarios.

Course Outcome: After completion of the course, the student will be able to
1. Understand the need, evolution and design issues of various categories of operating systems.
2. Apply different CPU scheduling algorithms and deadlock handling methods.
3. Analyze the principles of concurrency control and process synchronization problem.
4. Analyze various memory management techniques for efficient memory allocation.
5. Apply the concept of various I/O management, Disk scheduling and file system.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	1	1	1	-	-	-	-	2
CO2	3	3	3	3	3	1	-	-	-	-	2
CO3	3	3	2	3	3	2	-	-	-	-	2
CO4	3	3	2	3	3	2	-	-	1	-	2
CO5	3	3	2	2	2	2	-	-	1	-	2

Unit 1	**Introduction of Operating System**	**09 hours**

Introduction: Operating system Components and its services, Classification of Operating systems- Batch system, Time sharing, Real Time System, Multiprocessor Systems, Multiuser Systems, Multiprocess Systems, Multithreaded Systems, Operating System Structure- Layered structure, Reentrant Kernels, Monolithic and Microkernel Systems. System Calls, Elementary Linux commands and Shell Scripting.

Unit 2	**Process Scheduling and Resource Management**	**09 hours**

Introduction to Process: Process States, State Transition Diagram, Schedulers, Process Control Block (PCB), Threads and their management,

CPU Scheduling: Concepts, Performance Criteria, Scheduling Algorithms.

Deadlock: System model, Deadlock characterization, Prevention, Avoidance and detection, Recovery from deadlock.

Unit 3	**Concurrent Processes**	**09 hours**

Concurrent Processes: Principle of Concurrency, Critical Section Problem, Mutual Exclusion, Dekker's solution, Peterson's solution, Semaphores, Monitors, Test and Set operation; Classical Problem in Concurrency- Producer / Consumer Problem, Reader Writer Problem, Dining Philosopher Problem, Sleeping Barber Problem; Inter Process Communication models (IPC).

Unit 4	**Memory Management**	**09 hours**

Memory Management: Basic bare machine, Resident monitor, Multiprogramming with fixed partitions and variable partitions, Protection schemes, Paging, Segmentation, Paged segmentation, Virtual memory concepts, Demand paging, Performance of demand paging, Page replacement algorithms, Thrashing.

Unit 5	**I/O Management and Disk Scheduling:**	**09 hours**

File Systems and I/O Management of Windows and Linux.

Disk Scheduling: Disk storage structure and disk scheduling algorithms, RAID.

Case Study: Introduction to Android and Mac Operating System, The Evolution of Mobile Operating Systems: iOS vs. Android

Total Lecture Hours **45 hours**

Text Book:

1. Silberschatz, Galvin and Gagne, "Operating Systems Concepts", Wiley
2. Harvey M Dietel, " An Introduction to Operating System", Pearson Education
3. William Stallings, "Operating Systems: Internals and Design Principles ", 6th Edition, Pearson Education

Reference Book:

1. Sibsankar Halder and Alex A Aravind, "Operating Systems", Pearson Education
2. D M Dhamdhere, "Operating Systems : A Concept based Approach", 2nd Edition, TMH.
3. Andrew S. Tanenbaum and Herbert Bros, Modern Operating Systems (4thEdition), Pearson

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
60	15	75	150
75			

Course Code: MA105L	Course Name: Probability & Statistics	L	T	P	C
		3	0	0	3

Pre-requisite: NIL

Course Objectives:

1. To familiarize the graduate engineers with the concept of Statistics and Probability.
2. It aims to analyze the practical/ real life problems and solve them in scientific manner.

Course Outcome: After completion of the course, the student will be able to

1. Employ the concept of measure central tendency and regression analysis.
2. Apply knowledge of probability on distribution function.
3. Apply the concept of probability density function and normal distribution.
4. Apply the concept of random variable and time series.
5. Employ the knowledge of hypothesis by means of Chi-square and ANOVA test.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	2	2	-	-	-	1	-	-	-	1
CO2	2	2	2	-	-	-	1	-	-	-	1
CO3	2	2	1	-	-	-	1	-	-	-	1
CO4	2	2	1	-	-	-	1	-	-	-	1
CO5	2	2	2	-	-	-	1	-	-	-	1

Unit	Title	Hours
Unit 1	Basic Statistics	9 hours
	Introduction to Descriptive Statistics, Measure of Central Tendency, Histogram in sampling, Method of least square (basic concept), Fitting of Straight line and exponential curve, Correlation, Rank correlation and Regression Analysis.	
Unit 2	Probability I	9 hours
	Probability, Law of total Probability, Conditional Probability, Baye's Theorem, Discrete Random Variable, Probability Mass function. Binomial Distribution, Poisson Distribution., Introduction to confusion matrix.	
Unit 3	Probability II	9 hours
	Continuous Random Variable, Probability density function, Properties of Probability density function, Expectation and variance, Normal Distribution and its applications.	
Unit 4	Bivariate Random Variable and Time Series	9 hours
	Introduction to two-dimensional random variable, Joint probability density function and its properties, Marginal probability distribution. Introduction to Time series, component of time series, Measure of trend (Graphic method, method of Averages).	
Unit 5	Sampling Theory	9 hours
	Introduction to Inferential Statistics, Testing of Hypothesis: Introduction, Sampling Theory (Small and Large), Hypothesis, Null hypothesis, Alternative hypothesis, Testing a Hypothesis, Level of significance, Confidence limits, t-test, Chi-square test, one way analysis of variance (ANOVA).	
	Total Lecture Hours	45 hours

Textbook:

1. B. V. Ramana, Higher Engineering Mathematics, McGraw-Hill Publishing Company Ltd., 2017
2. B. S. Grewal, Higher Engineering Mathematics, Khanna Publisher, 2017.
3. R K. Jain & S R K. Iyenger, Advance Engineering Mathematics, Narosa Publishing House 2002.
4. S. C. Gupta & V. K. Kapoor, Fundamental of Mathematical Statistics, Sultan Chand & Sons.

Reference Books:

1. Seymour Lipschutz, John Schiller, Introduction to Probability and Statistics, McGraw Hill
2. Peter V. O'Neil, Advance Engineering Mathematics, Thomson (Cengage) Learning, 2007.
3. TKV Iyenger, B. Krishna Gandhi, S. Ranganatham, MVSN Prasad, Probability and Statistics (S. Chand Publishing House).
4. E. Kreyszig, Advance Engineering Mathematics, John Wiley & Sons, 2005.

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
60	15	75	150
75			

Course Code: HS109L	Course Name: Constitution of India	L	T	P	C
		2	0	0	NC

Pre-requisite: NIL

Course Objectives:

1. To acquaint the students with legacies of constitutional development in India and help those to understand the most diversified legal document of India and philosophy behind it.
2. To make students aware of the theoretical and functional aspects of the Indian Parliamentary System.
3. To channelize students' thinking towards basic understanding of the legal concepts and its implications for engineers.
4. To learn procedure and effects of emergency, composition and activities of election commission and amendment procedure.

Course Outcome: After completion of the course, the student will be able to

1. Understand basic features and modalities about Indian constitution.
2. Clarify the functioning of Indian parliamentary system at the center and state level.
3. Understand the aspects of Indian Legal System and its related bodies.
4. Apply different laws and regulations related to engineering practices.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	-	2	-	-	-	2
CO2	-	-	-	-	-	-	1	1	-	-	2
CO3	-	-	-	-	-	-	1	1	-	1	2
CO4	-	-	-	-	-	-	2	1	1	1	2

Unit 1 — Basic Information about Indian Constitution — 07 hours

Meaning of the constitution law and constitutionalism, Historical Background of the Constituent Assembly, Government of India Act of 1935 and Indian Independence Act of 1947, Enforcement of the Constitution, Indian Constitution and its Salient Features, The Preamble of the Constitution, Fundamental Rights, Fundamental Duties, Directive Principles of State Policy, Parliamentary System, Federal System, Centre-State Relations, Amendment of the Constitutional Powers and Procedure, The historical perspectives of the constitutional amendments in India, Emergency Provisions: National Emergency, President Rule, Financial Emergency, and Local Self Government – Constitutional Scheme in India.

Unit 2 — Union Executive and State Executive: — 08 hours

Powers of Indian Parliament Functions of Rajya Sabha, Functions of Lok Sabha, Powers and Functions of the President, Comparison of powers of Indian President with the United States, Powers and Functions of the Prime Minister, Judiciary – The Independence of the Supreme Court, Appointment of Judges, Judicial Review, Public Interest Litigation, Judicial Activism, Lok Pal, Lok Ayukta, The Lokpal and Lok ayuktas Act 2013, State Executives – Powers and Functions of the Governor, Powers and Functions of the Chief Minister, Functions of State Cabinet, Functions of State Legislature, Functions of High Court and Subordinate Courts.

Unit 3 — Basic Information about Legal System — 08 hours

The Legal System: Sources of Law and the Court Structure: Enacted law -Acts of Parliament are of primary legislation, Common Law or Case law, Principles taken from decisions of judges constitute binding legal rules. The Court System in India and Foreign Courtiers (District Court, District Consumer Forum, Tribunals, High Courts, Supreme Court). Arbitration: As an alternative to resolving disputes in the normal courts, parties who are in dispute can agree that this will instead be referred to arbitration. Contract law, Tort, Law at workplace.

Unit 4 — Election provisions, Emergency provisions, Amendment of the constitution — 07 hours

Election Commission of India-composition, powers and functions and electoral process. Types of emergency-grounds, procedure, duration and effects. Amendment of the constitution- meaning, procedure and limitations

Total Lecture Hours — 30 hours

Textbook:

1. Brij Kishore Sharma: Introduction to the Indian Constitution, 8th Edition, PHI Learning Pvt. Ltd.
2. Granville Austin: The Indian Constitution: Cornerstone of a Nation (Classic Reissue), Oxford University Press.
3. S.G Subramanian: Indian Constitution and Indian Polity, 2nd Edition, Pearson Education 2020.
4. Subhash C. Kashyap: Our Constitution: An Introduction to India's Constitution and constitutional Law, NBT, 2018.
5. Madhav Khosla: The Indian Constitution, Oxford University Press.
6. PM Bakshi: The Constitution of India, Latest Edition, Universal Law Publishing.
7. V.K. Ahuja: Law Relating to Intellectual Property Rights (2007)
8. Suresh T. Viswanathan: The Indian Cyber Laws, Bharat Law House, New Delhi-88
9. P. Narayan: Intellectual Property Law, Eastern Law House, New Delhi

10. Executive programme study material Company Law, Module II, by ICSI (The Institute of Companies Secretaries of India) (Only relevant sections i.e., Study 1, 4 and 36).https://www.icsi.edu/media/webmodules/publications/Company%20Law.pdf
11. Handbook on e-Governance Project Lifecycle, Department of Electronics & Information Technology, Government of India, https://www.meity.gov.in/writereaddata/files/e- Governance_Project_Lifecycle_Participant_Handbook-5Day_CourseV1_20412.pdf
12. Companies Act, 2013 Key highlights and analysis by PWC. https://www.pwc.in/assets/pdfs/publications/2013/companies-act-2013-key-highlights- and-analysis.pdf.

Reference Books:

1. Keshavanand Bharati V. State of Kerala, AIR 1973 SC 1461.
2. Maneka Gandhi V. Union of India AIR, 1978 SC 597.
3. S.R. Bammai V. Union of India, AIR 1994 SC 1918.
4. Kuldip Nayyar V. Union of India, AIR 2006 SC312.
5. A.D.M. Jabalpur V. ShivkantShakla, AIR 1976 SC1207.
6. Remshwar Prasad V. Union of India, AIR 2006 SC980.
7. Keshav Singh in re, AIR 1965 SC 745.
8. Union of India V. Talsiram, AIR 1985 SC 1416.
9. Atiabari Tea Estate Co.V. State of Assam, AIR 1961SC232.
10. SBP & Co. Vs. Patel Engg. Ltd. 2005 (8) SCC 618.
11. Krishna Bhagya Jala Nigam Ltd. Vs. G. Arischandra Reddy (2007) 2 SCC 720.
12. Oil & Natural Gas Corporation Vs. Saw Pipes Ltd. 2003 (4) SCALE 92 – 185.
13. Information Technology Act, 2000 with latest amendments. Compare this with GDPR of Europe
14. RTI Act 2005 with latest amendments.
15. Information Technology Rules, 2000
16. Cyber Regulation Appellate Tribunal Rules, 2000
17. RSTV debates on corporate law, IPR and patent issues
18. NPTEL lectures on IPR and patent rights
19. Episodes of 10 -part mini TV series "Samvidhan: The Making of Constitution of India" by RSTV.

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
25	-	**25**	**-**
25			

Course Code: HS110L	**Course Name: Aptitude-1**	**L**	**T**	**P**	**C**
		1	**0**	**0**	**1**

Pre-requisite: NIL

Course Objectives:
1. To provide adequate exposure to the students regarding the use of aptitude tests in the recruitment process and competitive examinations.
2. To improve the logical & numerical ability of the students.

Course Outcome: After completion of the course, the student will be able to
1. Understand the underlying patterns and logical principles involved in aptitude and reasoning problems.
2. Apply analytical and reasoning techniques to solve arrangement and information-based problem scenarios.
3. Analyze relationships, conditions, and constraints in complex reasoning situations to draw logical conclusions.
4. Evaluate arguments, assumptions, and alternative solution approaches to make justified decisions in reasoning-based contexts.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	1	-	1	-	1	-	-	-	-	1
CO2	1	1	-	1	-	2	-	-	-	-	1
CO3	1	1	-	1	-	1	-	-	-	-	2
CO4	1	1	-	2	-	1	-	-	-	-	2

Unit 1	**Series, Coding and Encoding**	**04 hours**
Importance and overview of Quantitative Aptitude and Logical Reasoning, Number Series, Letter Series, Analogies, Coding and decoding.		
Unit 2	**Data Arrangement**	**04 hours**
Ranking and Order, Direction Sense, Linear and Circular sitting arrangement.		
Unit 3	**Blood Relation and Puzzles**	**03 hours**
Basic concepts, definition and terminology related to blood relationships, Conversation-based blood relationships, Family Tree-based problems, Coded relationships and related puzzles.		
Unit 4	**Critical and Non-Verbal Reasoning**	**04 hours**
Statement arguments, course of action, classification and grouping of images, Figure series, Mirror image, Water image, Paper cutting, Paper folding, Embedded figures.		
	Total Lecture Hours	**15 hours**

Textbook:

1. A Modern Approach to Verbal & Non-Verbal Reasoning" by R.S. Aggarwal, S. Chand Publication.
2. https://www.geeksforgeeks.org/most-important-aptitude-topics-for-placements/

Reference Books:

1. "How to Prepare for Logical Reasoning for the CAT" by Arun Sharma, TMH Publication
2. https://www.indiabix.com/logical-reasoning/questions-and-answers/
3. https://testbook.com/placement-aptitude/test-series

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
-	25	25	50
25			

Course Code: HS111L	Course Name: Soft Skill Essentials-1	L	T	P	C
		1	0	0	NC

Pre-requisite:

- Communication skills subject in first year.
- Prior exposure to basic communication concepts (like verbal/non-verbal communication and listening skills)

Course Objectives:
To develop students' communication, presentation, and interpersonal skills through interactive activities, elevating confidence and professionalism for academic and workplace success.

Course Outcome: After completion of the course, the student will be able to

1. Demonstrate improved self-awareness and communication skills through structured presentations and vocabulary-building activities.
2. Apply effective verbal communication techniques, including pronunciation and elevator pitch delivery, to express ideas clearly and confidently.
3. Develop effective presentation skills that displays good non-verbal communication as well.
4. Exhibit professional behaviour, grooming, and teamwork skills in group discussions, interviews, and workplace-related role plays.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	-	-	2	3	-	2
CO2	-	-	-	-	-	-	-	2	3	-	2
CO3	-	-	-	-	-	1	-	3	3	-	2
CO4	-	-	-	-	-	1	-	2	3	-	2

Unit	Title	Hours
Unit 1	Foundation of Communication and Self-Awareness	05 hours
British Council-English Score Test, Team Presentations on Change Management Models, Presentations on Personality Profiling for professional growth		
Unit 2	Professional Expression and Digital Literacy	04 hours
Pronunciation Drill 1 & 2, Elevator Pitch Practice Session 1 & 2		
Unit 3	Professional and Workspace Readiness	06 hours
Professional Grooming and Etiquette, Group Discussion (General Topics), Panel Discussion on workplace scenarios using caselets		
	Total Lecture Hours	15 hours

Textbook:

1. www.mindtools.com
2. https://englishonline.britishcouncil.org/
3. www.toastmasters.org
4. https://www.futurelearn.com/
5. English Score Test
6. Duo Lingo Test

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
-	25	25	50
25			

Course Code: CS336B	Course Name: Object Oriented Programming with Java	L	T	P	C
		3	0	2	4

Pre-requisite: NIL

Course Objectives:

1. To familiarize students with the basic and advance Java Programming Language.
2. To learn modern tools to develop java-based web applications.

Course Outcome: After completion of the course, the student will be able to

1. Implement core Java concepts that model real world entities.
2. Apply a collection framework to build modular java programs
3. Develop programs based on new java features.
4. Implement JDBC and Spring Boot using Spring Framework concepts.
5. Implement Spring Boot using Spring Framework concepts.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	3	2	2	2	-	-	-	-	-	2
CO2	2	3	2	2	2	-	-	-	-	-	2
CO3	2	3	2	2	2	-	-	-	-	-	3
CO4	2	2	2	2	3	-	-	-	-	2	3
CO5	2	3	3	3	2	-	-	-	-	2	3

Unit 1 | Java Basics | 15 hours

Class, Object, Constructors, Methods, Access Specifies, Static Members, Final Members, Abstraction, Inheritance, Encapsulation, Polymorphism, Interface, Exceptions: Use of try, catch, finally, throw, throws, In built and User Defined Exceptions, Checked and Un-Checked Exceptions. Thread, Thread Life Cycle, Creating Threads, Thread Priorities, Synchronization.

Unit 2 | Java Collections | 15 hours

Collection in Java, Collection Framework in Java, Hierarchy of Collection Framework, Iterator Interface, Collection Interface, List Interface, ArrayList, LinkedList, Vector, Stack, Queue Interface, Set Interface, HashSet, LinkedHashSet, Sorted Set Interface, TreeSet, Map Interface, HashMap Class, Linked HashMap Class, TreeMap Class, Hashtable Class, Sorting, Comparable Interface, Comparator Interface, Properties Class in Java.

Unit 3 | Advance Java Features | 15 hours

Functional Interfaces, Lambda Expression, Method References, Stream API, Default Methods, Static Method, For Each Method, Try-with-resources, Java Module System, Diamond Syntax with Inner Anonymous Class, Local Variable Type Inference, Switch Expressions, Yield Keyword, Text Blocks, Records, Sealed Classes.

Unit 4 | Advance Java: Spring | 15 hours

Java Database Connectivity (JDBC): JDBC Drivers, JDBC with CRUD operations using MySQL; MVC, Spring Core Basics, Spring Dependency Injection concepts, Spring Inversion of Control, AOP, Bean Scopes- Singleton, Prototype, Request, Session, Application, Web Socket, Auto wiring, Annotations, Life Cycle Call backs, Bean Configuration styles.

Unit 5 | Advance Java: Spring boot | 15 hours

Spring Boot Build Systems, Spring Boot Code Structure, Spring Boot Runners, Logger, Building Restful Web Services, Rest Controller, Request Mapping, Request Body, Path, Variable, Request Parameter, GET, POST, PUT, DELETE APIs, Build Web Applications.

Total Lecture Hours | 75 hours

Textbook:

1. Herbert Schildt, "Java The complete reference", McGraw Hill Education
2. Steven Holzner, "Java Black Book", Dreamtech.
3. Balagurusamy E, "Programming in Java", McGraw Hill
4. Java: A Beginner's Guide by Herbert Schildt, Oracle Press
5. Greg L. Turnquist "Learning Spring Boot 2.0 - Second Edition", Packt Publication

Reference Books:

1. Kathey Sierra, "Head First Java", O'Reilly
2. AJ Henley Jr (Author), Dave Wolf, "Introduction to Java Spring Boot: Learning by Coding", Independently Published
3. Craig Walls, "Spring Boot in Action" Manning Publication

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
80	20	100	200
100			

Course Code: CS302B	Course Name: Advanced Data Structure	L	T	P	C
		3	0	2	4

Pre-requisite: The course requires a background in mathematics and strong programming skills, along with fundamental knowledge of data structures and algorithms.

Course Objectives:

1. This course develops a deep understanding of advanced data structures for efficient problem-solving.
2. Students will analyze, implement, and apply complex structures like trees, heaps, graphs, and hashing.

Course Outcome: After completion of the course, the student will be able to

1. Analyze algorithms using asymptotic notations and apply the Divide & Conquer technique for efficient problem-solving.
2. Implement hashing techniques with collision resolution strategies for optimized searching and storage.
3. Construct and manipulate tree-based data structures, including BST, AVL, Red-Black, B-Trees, and Huffman coding.
4. Apply graph representations and traversal techniques like BFS, DFS, Connected Components, and Topological Sorting.
5. Utilize advanced data structures such as Binomial Heaps, Tries, Skip Lists, and Splay Trees for efficient data processing.

CO-PO Mapping (Scale1:Low, 2:Medium,3:High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2	-	-	1	-	-	1	2
CO2	3	3	3	2	-	-	1	-	-	1	2
CO3	3	3	3	2	-	-	1	-	-	1	2
CO4	3	3	3	2	-	-	1	-	-	1	2
CO5	3	3	3	2	-	-	1	-	-	1	2

Unit 1 | Introduction & Divide and Conquer Technique | 15 hours

Introduction: Algorithms, Analyzing algorithms - Orders of Magnitude (Asymptotic notations), Growth of functions (constant, logarithmic, linear, polynomial, exponential). Divide & Conquer Technique: Binary search, Merge sort, Quick sort, Randomized quick sort, Efficiency analysis of sorting.

Problem solving:

1. Write a program to implement the Binary Search algorithm using Divide and Conquer technique on a sorted array. Accept an element from the user and search for it in the array. Display the number of comparisons made.
2. Given an array of integers nums sorted in non-decreasing order, and an integer target, find the starting and ending position of the target value. If the target is not found in the array, return [-1, -1]. Your solution must run in O (log n) time complexity.
3. Write a program to implement Merge Sort using Divide and Conquer. Accept n elements from the user, sort them, and display the sorted output along with the number of recursive calls and comparisons made.
4. Given an array of integers, count the number of inversions in the array. An inversion is a pair of elements (i, j) such that: i < j and arr[i] > arr[j], Your task is to write a program that returns the total number of such inversion pairs in the array.
5. Write a program to implement Quick Sort using the last element as the pivot. Accept an unsorted array, sort it, and count the number of comparisons and recursive calls.
6. Write a program to implement Randomized Quick Sort where the pivot is chosen randomly. Compare the number of comparisons and execution time with the regular Quick Sort.
7. Given an integer array nums, find the contiguous subarray (containing at least one number) which has the largest sum, and return its sum using the Divide and Conquer approach.
8. Given an integer array nums and an integer k, return the kth largest element in the array. Note that it is the kth largest element in sorted order, not the kth distinct element.

Unit 2 | Hashing | 15 hours

Hashing & Searching Techniques: Introduction, Hash table, Hash function, Collision resolution technique: Open hashing (Separate Chaining), Closed Hashing - Linear Probing, Quadratic probing, Double Hashing.

Problem solving:

1. Write a program to implement a hash table using the open hashing (separate chaining) method. Use an array of linked lists to handle collisions. Support insertion, deletion, and search operations. Test your implementation by inserting multiple keys and observing how collisions are handled using chaining.
2. Given an array of integers, find the frequency of each distinct element using hashing and return a map of element to its count.
3. Given an array and an integer *k*, check whether the array contains duplicate elements within a distance *k* from each other using hashing.
4. Given an array and a value *sum*, determine whether any two numbers in the array add up to the sum using a hash set for efficient search.
5. Given an array and a window size *k*, print the count of distinct elements in every contiguous window of size *k* using hashing.
6. Given an integer array nums and an integer k, return the k most frequent elements. You may return the answer in any order.
7. Given an array of strings strs, group the anagrams together. You can return the answer in any order. Two strings are anagrams of each other if they contain the same characters in the same counts, but possibly in a different order.
8. Given an array of integers nums and an integer target, return indices of the two numbers such that they add up to the target.

Unit 3 | Trees | 15 hours

Tree: Binary Trees, Binary Search Trees (BST), Threaded Binary Trees, Huffman coding, Balanced Trees: AVL Tree & its operation (insertion, deletion).

Problem solving:

1. Write a program to implement a binary tree with basic operations such as insertion (level-order), in-order, pre-order, and post-order traversals. Test the tree with a set of integers and display the output of each traversal method.
2. Implement a binary search tree (BST) with functions to insert, delete, and search a node. Perform and display in-order traversal after each operation to show the structure of the BST.

3. Write a program to build a Huffman tree for a given set of characters and their frequencies, and generate Huffman codes for each character. Display the codes and the structure of the tree used in encoding.
4. Create a program to implement an AVL Tree that supports insertion of elements. Display the tree structure (in-order traversal) after each insertion and show the rebalancing operations (rotations) performed.
5. Given the root of a binary tree, find its maximum depth (also called height). The maximum depth is the number of nodes along the longest path from the root node down to the farthest leaf node**.**
6. Given two integer arrays, preorder and inorder, where: preorder is the Preorder Traversal of a binary tree, inorder is the Inorder Traversal of the same tree, Construct and return the binary tree.
7. Given the root of a binary tree, return the length of the diameter of the tree. The diameter of a binary tree is the longest path between any two nodes in the tree. This path may or may not pass through the root.

Note: The length of a path between two nodes is measured by the number of edges between them.

8. Given a binary tree and two nodes p and q, find their lowest common ancestor (LCA). The LCA of two nodes p and q is the lowest node in the tree that has both p and q as descendants (where a node can be a descendant of itself).

Unit 4	**Graph**	**15 hours**

Graphs: Terminology, Sequential and linked Representations of Graphs: Adjacency Matrices, Adjacency List, Adjacency Multi list, Graph Traversal: Depth First Search and Breadth First Search, Connected Component, Topological sort.

Problem solving:

1. Implement a graph using an adjacency matrix and provide functions to:
   - Add/remove vertices and edges.
   - Check if an edge exists between two vertices.

   Print the matrix.
2. Implement a graph using an adjacency list (linked lists or dynamic arrays) and support:
   - Edge insertion/deletion.
   - Display neighbors of a vertex.
3. Implement DFS to traverse a graph from a given start vertex and print the visit order. Handle disconnected graphs.
4. Implement BFS to traverse a graph and print vertices level by level.
5. Find all connected components in an undirected graph using DFS/BFS.
6. Perform topological sort on a directed acyclic graph (DAG) using Kahn's algorithm (BFS-based).
7. Implement DFS-based topological sort for a DAG.
8. Determine if a graph is bipartite (can be colored with 2 colors such that no adjacent nodes have the same color).
9. Given an m x n 2D binary grid grid which represents a map of '1's (land) and '0's (water), return the number of islands. An island is a group of adjacent '1's connected horizontally or vertically (not diagonally). You may assume all four edges of the grid are surrounded by water.

Unit 5	**Advanced Data Structure**	**15 hours**

Advanced Data Structure: Binomial Heap, Operations on Binomial Heap, Trie data structures, Priority Queue, Disjoint data structure.

Problem solving:

1. Implement a binomial heap and write a function to merge two binomial heaps.
2. Extract the minimum key from a binomial heap and rebalance the heap.
3. Given an undirected graph, detect if it contains a cycle using Union-Find.
4. Design a Trie (prefix tree) data structure that supports the following operations:
   - insert(word): Inserts the string word into the trie.
   - search(word): Returns True if the string word is in the trie (i.e., it was inserted before), and False otherwise.
5. Design a program that simulates a job scheduler using a Priority Queue, where each job has a priority and a name. It should support three operations: (1) Add a Job with a given priority and name, (2) Execute the highest priority job, resolving ties by insertion order, and (3) Display all pending jobs in descending priority order, maintaining insertion order for equal priorities. The implementation must use an efficient priority queue structure like a heap with a tie-breaker mechanism for insertion order.

Total Lecture Hours	**75 hours**

Textbook:

1. Cormen, Leiserson, Rivest, and Stein, Introduction to Algorithms (3 ed.), MIT Press, 2009. ISBN 978-0262033848.
2. Horowitz and Sahani, "Fundamentals of Data Structures", Galgotia Publications Pvt Ltd Delhi India.
3. Lipschutz, "Data Structures" Schaum's Outline Series, Tata McGraw-hill Education (India) Pvt. Ltd.
4. Reema Thareja, "Data Structure Using C" Oxford Higher Education.
5. AK Sharma, "Data Structure Using C", Pearson Education India.

Reference Books:

1. V. Aho, J. E. Hopcroft, and J. D. Ullman, Data Structures and Algorithms (1 ed.), Pearson, 1983. ISBN 978-0201000238.
2. Dasgupta, Papadimitrou and Vazirani, Algorithms (3 ed.), McGraw-Hill Education, 2006. ISBN 978-0073523408.
3. Horowitz, Sahni, and Rajasekaran, Computer Algorithms (2 ed.), Silicon Press, 2007. ISBN 978-0929306414.
4. Kleinberg and Tardos, Algorithm Design (1 ed.), Pearson, 2005. ISBN 978-0321295354.

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
60	15	**75**	**150**
75			

Course Code: CS205B	Course Name: Artificial Intelligence and Its Applications	L	T	P	C
		3	0	2	4

Pre-requisite: Python Programming

Course Objectives:

1. To provide students with a foundational understanding of Artificial Intelligence (AI) concepts, tools, and practical implementations, including search algorithms, knowledge representation, and ethical considerations.
2. To equip students with hands-on experience in designing and implementing agentic AI systems, genetic algorithms, and their real-world applications across diverse domains such as healthcare, finance, and robotics.

Course Outcome: After completion of the course, the student will be able to

1. Demonstrate foundational understanding of Artificial Intelligence, its key domains, tools, and ethical implications by implementing logic-based systems and classical search algorithms
2. Design and simulate intelligent agents and multi-agent systems in diverse environments, incorporating concepts of decision-making, utility, and communication strategies.
3. Apply reinforcement learning techniques to develop learning agents and multi-agent environments, and visualize their behavior in real-world-inspired simulations and games.
4. Construct and optimize Genetic Algorithm-based solutions to solve complex real-world problems such as scheduling, path planning, and feature selection, while exploring hybrid and advanced evolutionary strategies.
5. Analyze the societal impacts, ethical risks, biases, and economic challenges of AI systems through real-world case studies.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	3	3	1	-	-	-	-	2
CO2	2	3	3	2	3	1	-	-	-	-	2
CO3	2	3	3	2	2	1	-	-	-	-	3
CO4	1	2	2	3	2	3	-	-	-	-	3
CO5	-	2	-	2	-	2	3	2	2	-	2

Unit 1 | Practical Foundations of Artificial Intelligence | 18 hours

Introduction to AI and its Tools & Libraries- Definition, History, and Evolution of AI, Key Domains: Machine Learning, Deep Learning, NLP, Computer Vision, AI Applications in Healthcare, Finance, and Robotics Python for AI: NumPy, Pandas, and Scikit-learn Introduction to TensorFlow/PyTorch Using Jupyter Notebooks for Experiments.

Search Algorithms with Implementation- Uninformed Search Strategies: BFS, DFS, Informed Search Strategies: A*, Greedy Best-First Search Constraint Satisfaction Problems (CSPs) BFS & DFS Implementation in Python Developing a Maze Solver using Search Algorithms Implementing A* Algorithm for Path Planning.

Knowledge Representation & Ethics- Propositional Logic and First-Order Logic, Knowledge Graphs and Ontologies Responsible AI Practices, Challenges of Fairness, Transparency, and Explainability Implementing Logic-Based Systems with Python Libraries Building a Rule-Based Expert System.

Mini Projects: Implement a Tic-Tac-Toe AI using Minimax Algorithm Create a Pathfinding Visualizer for BFS/DFS/A*

Unit 2 | Foundations of Agentic AI and Intelligent Agents | 19 hours

Introduction to Agents and Agent Environments- Definition and Characteristics of Agents Types of Agents: Reflex, Goal-Based, Utility-Based Agent Environments: Fully vs. Partially Observable.

Designing and Implementing Simple Agents in Python- Build Reflex and Goal-Based Agents Smart Vacuum Cleaner Simulation Self-Navigating Agent in a Virtual Grid.

Multi-Agent Systems and Distributed AI- Overview of Multi-Agent Systems (MAS) Coordination and Communication in MAS Real-World Applications: Smart Grids, Collaborative Robot.

Decision-Making and Agent Intelligence- Introduction to Markov Decision Processes (MDPs) Basics of Game Theory and Nash Equilibrium Designing Rule-Based Decision Logic.

Mini Project: Build a Reflex Agent for Maze Navigation

Unit 3 | Learning Agents and Practical Agentic AI Systems | 19 hours

Learning Agents and Reinforcement Learning- Learning Agents: Characteristics and Architectures Reinforcement Learning Fundamentals Introduction to OpenAI Gym.

Implementing RL Algorithms- Q-Learning: Concepts and Python Implementation Deep Q-Networks (DQN): Architecture and Training Visualizing Agent Learning Progress.

Agentic AI for Games and Simulation- Develop AI Agent for Snake or Flappy Bird Case Study: Self-Learning Bot in a Custom Game Reward Shaping and Environment Design.

Advanced Multi-Agent Learning and Simulations- Cooperative and Competitive Agents Traffic Control Simulation using Multi-Agent Systems, Reinforcement Learning in Multi-Agent Scenarios.

Mini Projects -Solve Frozen Lake Environment using Q-Learning

Unit 4 | Genetic Algorithms & Applications | 19 hours

Fundamentals of Evolutionary Computation- Introduction to Evolutionary Algorithms, Darwinian Principles: Selection, Mutation, Crossover. Genetic Algorithm Workflow and Structure.

Core Components of Genetic Algorithms- Chromosome Representation Techniques, Fitness Functions and Selection Strategies, Crossover Techniques (One-point, Two-point, Uniform), Mutation Methods.

Implementing Genetic Algorithms in Python- Writing a Basic GA from Scratch, Custom Fitness Function Design, Visualizing GA Evolution with Matplotlib.

Advanced Techniques and Hybridization- Elitism and Diversity Preservation, Parameter Tuning for Performance Optimization, Hybrid GAs: Integration with Local Search and Simulated Annealing, Introduction to Memetic Algorithms.

Practical Applications- Solving the Traveling Salesman Problem (TSP), Feature Selection in Machine Learning, Scheduling and Resource Allocation Problems, Autonomous Route Planning (e.g., for vehicles or drones)

Mini Project - Solve Optimal Timetable Scheduling for a University

Total Lecture Hours | **75 hours**

Online Resource: NPTEL's free AI Course by Prof. Deepak Khemani, IIT Madras

Textbook:

1. S. Russell and P. Norvig, "Artificial Intelligence: A Modern Approach‖, Prentice Hall, Third Edition, 2009.
2. I. Bratko, ―Prolog: Programming for Artificial Intelligence‖, Fourth edition, Addison-Wesley Educational Publishers
3. M. Tim Jones, ―Artificial Intelligence: A Systems Approach(Computer Science)‖, Jones and Bartlett Publishers, Inc.;First Edition, 2008
4. Nils J. Nilsson, ―The Quest for Artificial Intelligence‖, Cambridge University Press, 2009.
5. William F. Clocksin and Christopher S. Mellish,‖ Programming in Prolog: Using the ISO Standard‖, Fifth Edition,Springer, 2003.
6. Gerhard Weiss, ―Multi Agent Systems‖, Second Edition, MIT Press, 2013.
7. David L. Poole and Alan K. Mackworth, ―Artificial Intelligence: Foundations of Computational Agents‖, Cambridge University Press, 2010.

Reference Books:

1. Artificial Intelligence, Rich and Knight, (TMH)
2. Artificial Intelligence with Python" Prateek Joshi (Indian Author, Packt Publishing)

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
80	20	**100**	**200**
100			

Practical Courses Detail Syllabus – III Sem

Course Code: IT301P	Course Name: Database Systems Lab	L	T	P	C
		0	0	2	1

Pre-requisite: Concept of any programming Language

Course Objectives:

1. Develop Hands-on SQL Skills
2. Design and Normalize Databases
3. Work with PL/SQL and Advanced Database Concepts

Course Outcome: After completion of the lab, the student will be able to

1. Design Logical and Conceptual database schema for real life problem using ERD
2. Apply SQL to store, retrieve and manipulate data in relational database.
3. Apply PL/SQL to solve real world database management and automation task.
4. Create test database for the real world application using required database objects.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	2	3	1	3	1	1	2	2	1	2
CO2	3	3	2	1	3	1	1	1	1	1	2
CO3	3	3	2	1	3	1	1	1	2	1	2
CO4	3	2	3	1	3	-	-	-	-	-	2

List of Experiments:

1. **Creating Entity-Relationship Diagram using case tool (STAR UML)**

[Group formation and assigned project to submit a report at the end of the semester as project-based learning]

1	Insurance Database
2	Banking Enterprise Database
3	Supplier Database
4	Student Faculty Database
5	Airline Flight Database
6	Order Processing Database

7	Book dealer Database
8	Student Enrolment Database
9	Movie Database
10	College Database
11	Hostel Database
12	Library Database
13	Clinic Database
14	Medical Store Database
15	Exam Process Database

2. **Introduction to date types and database implementation using Create, insert, and Basic SQL SELECT statements.**
Create these two tables with following specifications and insert data in the table:

Name: Client_master

Attribute	**Data Type**	**Size**
Client_no	number	10
Client_Name	Varchar2	20
City	Varchar2	15
State	Varchar2	15
Pin	Number	6
Balance_due	Number	10,2

Data for Client master:

CLIENT_NO	**Client_NAME**	**CITY**	**STATE**	**PIN_CODE**	**BAL_DUE**
0001	Ivan	Bombay	Maharastra	400057	15000
0002	Vandura	Madras	Tamilnadu	980001	0
0003	Pramod	Bombay	Maharastra	400057	5000
0004	Basu	Bombay	Maharastra	400056	0
0005	Ravi	Delhi	Null	100001	2000
0006	Rukmini	Bombay	Maharastra	900050	0

Name: PRODUCT_MASTER

COLUMN	**DATA TYPE**	**Size**
PRODUCT_NO	VARCHAR2	6
DESCRIPTION	VARCHAR2	20
PROFIT%	NUMBER	10
QTY_ON_HAND	NUMBER	10
ORDER_LEVEL	NUMBER	10
SELL_PRICE	NUMBER	10
COST_PRICE	NUMBER	10

Data for Product_Master Table

Product_No	**Description**	**Profit %**	**Qty_on hand**	**Reorder_level**	**Sell price**	**Cost price**
P00001	1.44 floppies	5	100	20	525	500
P03453	Monitors	6	10	3	12000	11200
P06734	Mouse	5	20	5	1050	500
P07865	1.22 floppies	5	100	20	525	500
P07868	Keyboards	2	10	3	3150	3050
P07885	CD drive	2.5	10	3	5250	5100
P07965	540 HDD	4	10	3	8400	8000
P07975	1.44 Drive	5	10	3	1050	1000
P08865	1.22 Drive	5	2	3	1050	1000

Perform following queries on the above data:

(a) Find out the name of all the clients.
(b) Retrieve the list of names and cities of all the clients.
(c) List all the clients who are located in Bombay.
(d) Display the information for client no 0001 and 0002.
(e) Find the list of all clients who stay in city ‘Bombay’ or ‘Delhi’ or ‘Madras’.
(f) List the name, city, and state of clients not in state of ‘Maharashtra’

3. **To manipulate data in the existing tables**
Using the table client master and product master answer the following queries:
(a) Delete the record of Client no. 0001 from the Client master table.

(b) Change the city of Client no. 0005 to 'Bombay'.
(c) Change the balance due of Client no. 0002 to 1000.
(d) Find out the clients who stay in a city or state where second letter is a.
(e) Calculate the average balance due of all the clients.
(f) Change the selling price of 1.44 floppy drive to Rs. 1150.00.
(g) Count the number of products having price greater than or equal to 1500.

4. **To create, manage tables with constraints and alter the structure of tables.**
Create the tables with following specifications and constraints:

TABLE NAME: SALES_MASTER

ATTRIBUTE	DATA TYPE	SIZE	CONSTRAINT
SALESMAN_NO	VARCHAR2	6	PRIMARY KEY, FIRST LETTER IS 'S'
SALES_NAME	VARCHAR2	20	NOT NULL
ADDRESS	VARCHAR2	20	NOT NULL
CITY	VARCHAR2	20	---
STATE	VARCHAR2	20	---
PINCODE	NUMBER	6	---
SAL_AMT	NUMBER	8,2	NOT NULL, CAN'T BE ZERO
Tgt_to_get	NUMBER	6,2	NOT NULL, CAN'T BE ZERO
Ytd_sales	NUMBER	6,2	NOT NULL, CAN'T BE ZERO
Remark	VARCHAR2	30	

DATA OF SALES_MASTER:

Sales No.	Sales_Name	Address	City	Pincode	State	Salamt	Tgt_to_get	Ytd_sales	Remark
S00001	Kiran	A/14 worli	Bombay	400002	MAH	3000	100	50	Good
S00002	Manish	65, Nariman	Bombay	400001	MAH	3000	200	100	Good
S00003	Ravi	P-7, Bandra	Bombay	400032	MAH	3000	200	100	Good
S00004	Ashish	A/5 Juhu	Bombay	400044	MAH	3500	200	150	Good

TABLE NAME: SALES_ORDER

ATTRIBUTE	DATA TYPE	SIZE	CONSTRAINT
S_ORDER_No	VARCHAR2	6	PRIMARY KEY, FIRST LETTER IS 'O'
S_ORDER_DATE	DATE	---	---
CLIENT_NO	NUMBER	10	FOREIGN KEY FROM CLIENT MASTER
SALESMAN_NO	VARCHAR2	26	FOREIGN KEY FROM SALES MASTER
DELIVERY_TYPE	CHAR	1	P FOR PARTIAL AND F FOR FULL, DEFAULT IS F
BILLED_YN	CHAR	1	'Y' FOR YES AND 'N' FOR NO
DELIVERY_DATE	DATE	---	CAN'T BE LESS THAN S_ORDER_DATE
ORDER_STATUS	VARCHAR2	10	IN(IN-PROCESS, FULFILLED, BACK ORDER, CANCELLED)

DATA OF SALES_ORDER

S_order_no	S_order_date	Client no	Dely type	Bill yn	Salesman no	Delay date	Orderstatus
O19001	12-Jan-96	1	F	N	50001	20-Jan-96	IP
O19002	25-Jan-96	2	P	N	50002	27-Jan-96	C
O16865	18-Feb-96	3	F	Y	500003	20-Feb-96	F
O19003	03-Apr-96	1	F	Y	500001	07-Apr-96	F
O46866	20-May-96	4	P	N	500002	22-May-96	C
O10008	24-May-96	5	F	N	500004	26-May-96	IP

TABLE NAME: Sales_order_detail

Column	Dadatype	Size	Attributes
S_order_no	Varchar2	6	PK/FK references s_order_no of sales_order
Product_no	Varchar2	6	PK/FK references product_no of product_master
Qty_order	Number	8	
Qty_disp	Number	8	
Product_rate	Number	10,2	

Data for sale_order_detail

S_order_no	Product_no	Qty_order	Qty_disp	Product_rate
O19001	P00001	4	4	525
O19001	P07965	2	1	8400
O19001	P07885	2	1	5250
O19002	P00001	10	0	525
O46865	P07868	3	3	3150
O46865	P07885	10	10	5250
O19003	P00001	4	4	1050

O19003	P03453	2	2	1050
O46866	P06734	1	1	12000
O46866	P07965	1	0	8400
O10008	P07975	1	0	1050
O10008	P00001	10	5	525

(a) Make client_no primary key in client_master.
(b) Add new column phone_number inclient_master table.
(c) Add not null constraint in product master with columns : description, profit_percent, sellprice, costprice
(d) Change size of client_no field in client_master.
(e) Add check constraint to product_master such that sellprice is always greater than costprice.
(f) Select produc_no,description where profit percent is between 20 and 30 both inclusive

5. **To implement join concepts.**
(a) fiind out the product which has been sold to 'Ivan'.
(b) Find out the product and their quantities that will have to be delivered.
(c) Find out the names of clients who have purchased 'CD DRIVE'
(d) List the product_no and s_order_no of customers having qty ordered less than 5 from the order details table for the product '1.44 floppies'.
(e) Find the product and their quantities for the orders placed by 'Vandan' and 'Ivan'.
(f) Find the products and their quantities for the orders placed by client_no 'C00001"
(g) Find the order_no, Client_noanssalesman_no where a client has been received by more than one salesman.

6. **To aggregate data using group function and implement the concept of sub-queries.**
Perform following queries based on all 5 tables mentioned above:
(a) Print the description and total quantity sold for each product.
(b) Find the value of each product sold.
(c) Find out the products which have been sold to 'Ivan'.
(d) Find the names of clients who have 'CD Drive'.
(e) Find the products and their quantities for the orders placed by 'Vandana' and 'Ivan'
(f) Select product_no, total_qty_ordered for each product.
(g) Display the order number and day on which clients placed their order.
(h) Display the month and date when the order must be delivered.

To implement concept of sub-queries.
(a) Find the product_no and description of non moving products.
(b) Find the customer name, address, city and pincode for the client who has placed order no "019001".
(c) Find the client name who have placed order before the month of may 2006.
(d) Find out if product "1.44 Drive" is ordered by only client and print the client_no, name to whom it was soled.
(e) Find the name of client who have placed orders worth Rs. 10000 or more.
(f) Select the orders placed by "Rahul Desai".
(g) Select the name of person who are in Mr.Pradeep's department and who have also worked on inventory control system.
(h) Select all the clients and the slaesman in the city of Bombay.
(i) Select slaesman name in Bombay who has atleast one client located at Bombay.
(j) Select the product_no, description, qty_on-hand, cost_price of non_moving items in the product_master tab.

7. **To implement the concept of views and indexes**
(a) Create an index on the table client_master, field client_no.
(b) Create an index on the sales_order, fields_order_no.
(c) Create a composite index on the sales_order_details table for the columns_order_no. and product_no.
(d) Create view on salesman_master whose sal_amt is less than 3500.
(e) Create a view client_view on client_master and rename the columns as name, add1, add, city, pcode, state respectively.
(f) Select the client names from client_view who live in city 'Bombay'
(g) Drop the view client_view.

8. **To implement concept of PL/SQL**
(a) WAP in PL/SQL for addition of two numbers.
(b) WAP in PL/SQL for addition of 1 to 100 numbers.
(c) WAP in PL/SQL to inverse a number, eg. NUMBER 5639 when inverted must be display as output 9365.

9. **To implement concept of cursor**
Create a explicit cursor which updates the salary of an employee such that,
(a) If salary > 10000, then increase the salary by 15%
(b) If 5000<salary <10000, then increase the salary by 12%
(c) Otherwise, increase the salary by 25%.

10. **To create procedures and functions and triggers in Oracle.**

```
DECLARE
        x VARCHAR2(20);
BEGIN
        SELECT RTRIM (TO_CHAR(SYSDATE, 'DAY'), ' ') INTO x from DUAL;
        IF x = 'SUNDAY' THEN
        RAISE_APPLICATION_ERROR (-20001, 'Transaction is not allowed');
END IF;
END;
```

This package has one subprogram; procedure, update_sal.

```
CREATE OR REPLACE PACKAGE emp_pack
        IS
                PROCEDURE update_sal (eno IN NUMBER);
        END emp_pack;
CREATE OR REPLACE BODY emp_pack
        IS
          PROCEDURE update_sal (eno IN NUMBER)
                IS
                        x  EMP.Empno % type
                        y  EMP. Sal % type
                BEGIN
                SELECT empno, sal INTO x, y FROM emp WHERE empno=eno;
                        IF y> 3000 THEN
                                UPDATE EMP SET Sal= sal*1.1 WHERE Empno =eno;
                        ELSIF y between 2000 AND 3000 THEN
                                UPDATE EMP SET Sal= Sal*1.05 WHERE Empno =eno;
                        ELSE
                                UPDATE EMP SET Sal= Sal*1.03 WHERE Empno =eno;
                        END IF;
```

Calling a Package

Calling a package means actually referencing one of its elements. Following is the method for calling an element from a package.

```
DECLARE
      x VARCHAR2(20);
BEGIN
     emp_pack.update_sal (7633);
END;
```

Total Laboratory Hours: 30 hours

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
-	25	**25**	**50**
25			

Course Code: CS206P	Course Name: Operating Systems Lab	L	T	P	C
		0	0	2	3

Pre-requisite: Basic concept of C language and data structure

Course Objectives:
This course provides hands-on experience with operating system concepts, focusing on UNIX/Linux environments. After this course student will be able to apply the operating system algorithms to solve the real-life problems.

Course Outcome: After completion of the course, the student will be able to

1. Apply knowledge of basic UNIX System calls and Shell programming.
2. Implement various CPU scheduling algorithms and deadlock handling techniques.
3. Implement memory management, process synchronization techniques, page replacement techniques.
4. Implement various disk scheduling techniques.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	3	-	1	1	1	1	2
CO2	3	3	1	2	2	-	1	2	2	1	2
CO3	3	3	-	2	2	-	1	2	2	1	2
CO4	3	3	-	2	2	-	1	2	2	1	2

List Of Practical's (Indicative & Not Limited To):

1. Study of hardware and software requirements of different operating systems (UNIX, LINUX, WINDOWS, Android, IOS
2. Execute various UNIX system calls
3. Write a basic program on VI Editor in Linux.
4. Shell Scripting.
5. Implement FCFS CPU Scheduling Policy
6. Implement SJF CPU Scheduling Policy
7. Implement Priority CPU Scheduling Policy
8. Implement Round Robin CPU Scheduling Policy
9. Implementation of Banker's algorithm
10. Implement the solution for Bounded Buffer (producer-consumer) problem using inter process communication techniques-Semaphores
11. Implement the solutions for Readers-Writers problem using inter process communication technique – Semaphore
12. Implementation First Fit contiguous allocation technique
13. Implementation Best Fit contiguous allocation technique
14. Implementation Worst Fit contiguous allocation technique
15. Implement file storage allocation technique:
    -Contiguous (using array)
16. Implement file storage allocation technique:
    - Linked –list (using linked-list)
    -Indirect allocation (indexing)
17. Comparison of Disk Scheduling Algorithms.

Total Lecture Hours	**30 hours**

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
-	25	**25**	**50**
25			

Theory Courses Detail Syllabus – IV Sem

Course Code: CS401L	Course Name: Design & Analysis of Algorithm	L	T	P	C
		3	0	0	3

Pre-requisite: The course requires a strong foundation in mathematics, algorithmic problem-solving, and proficient programming skills.

Course Objectives:
1. Develop a strong understanding of algorithm design techniques for efficient problem-solving.
2. Analyze algorithm complexity and explore NP-completeness, approximation, and randomized algorithms.

Course Outcome: After completion of the course, the student will be able to
1. Analyse algorithm efficiency using recurrence relations and sorting techniques.
2. Solve optimization problems with Greedy algorithms (MST, Shortest Path).
3. Implement Dynamic Programming for Knapsack, Matrix Chain Multiplication, etc.
4. Apply Backtracking and Branch & Bound to combinatorial problems.
5. Understand NP-Completeness, Approximation, and Randomized algorithms.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	2	2	-	1	-	-	1	2
CO2	3	3	3	2	2	-	1	-	-	1	2
CO3	3	3	3	3	2	-	1	-	-	1	2
CO4	3	3	2	3	2	-	1	-	-	1	2
CO5	3	3	2	3	2	-	1	-	-	1	2

Unit 1 | Introduction & Sorting in Linear Time | 09 hours

Introduction: Algorithms, Algorithm Analysis, Recurrence Relations- substitution, Master's method, Change of Variables, shell sort, Tim sort, Linear time sorting – Counting sort, Radix sort, Bucket sort.

Unit 2 | Greedy | 09 hours

Greedy Algorithm: General Characteristics, Problem solving using Greedy Algorithm- Fractional Knapsack problem, Activity selection problem. Task scheduling with deadline and penalty problem, Minimum Spanning trees (Kruskal's algorithm, Prim's algorithm). Single source shortest paths - Dijkstra's and Bellman Ford algorithms.

Unit 3 | Dynamic Programming | 09 hours

Dynamic Programming: Introduction, The Principle of Optimality, Problem Solving using Dynamic Programming- optimal binary search trees, 0-1 Knapsack, Coin Change Problem, Assembly Line Scheduling, Longest Common Subsequence, Word break problem, Matrix chain multiplication, Rod cutting problem, All pair shortest paths – Warshal's and Floyd's algorithms, Resource allocation problem.

Unit 4 | Backtracking & Branch and Bound | 09 hours

Back Tracking: Introduction, Subset sum problem, Permutation, N-queen problem, Graph Coloring, Hamiltonian Cycles and Sum of subsets, analysis of these problems. Branch and Bound: Traveling salesman problem.

Unit 5 | String Matching, Approximation Algorithm | 09 hours

String Matching: Introduction, Naive string matching algorithm, Rabin-Karp algorithm, String Matching with finite automata, KMP (Knuth Morris Pratt) matching algorithm, Boyer Moore String matching.
Approximation algorithms: Travelling Salesman problem, Hamiltonian problem, Vertex Cover Problem. Introduction to NP-Completeness: The class P and NP, Polynomial reduction, NP Completeness Problem, NP-Hard Problems. Randomized Algorithm.

Total Lecture Hours | 45 hours

Textbook:
1. Cormen, Leiserson, Rivest, and Stein, Introduction to Algorithms (3 ed.), MIT Press, 2009. ISBN 978-0262033848.

Reference Books:
1. Dasgupta, Papadimitrou and Vazirani, Algorithms (3 ed.), McGraw-Hill Education, 2006. ISBN 978-0073523408.
2. Design and Analysis of Computer Algorithms by Aho, Hopcroft and Ullman, Pearson.
3. Horowitz, Sahni, and Rajasekaran, Computer Algorithms (2 ed.), Silicon Press, 2007. ISBN 978-0929306414.
4. Kleinberg and Tardos, Algorithm Design (1 ed.), Pearson, 2005. ISBN 978-0321295354.

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
60	15	75	150
75			

Course Code: IT302L	Course Name: Computer Networks	L	T	P	C
		3	0	0	3

Pre-requisite: Computer Fundamentals

Course Objectives:

The objective of this course is to provide insight about computer network concepts and to gain comprehensive knowledge of the layered communication architectures, its functionalities, key protocols as well as security issues related with each layer.

Course Outcome: After completion of the course, the student will be able to

1. Apply the knowledge of networking concepts and functionality of physical layer.
2. Explore the concept of elementary data link layer protocol to build a robust network.
3. Analyze the concept of routing and IP addressing in network layer.
4. Examine the usage and working of transport layer.
5. Determine the performance of different protocols used at application layer.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	-	-	-	-	-	-	-	2
CO2	3	3	-	-	2	-	-	-	-	-	2
CO3	3	3	-	-	2	-	-	-	-	2	2
CO4	3	3	-	-	-	-	-	-	-	-	2
CO5	3	3	-	-	-	-	-	-	-	-	2

Unit 1	**Introduction -Physical Layer**	**9 hours**

Introduction to computer networks: Network applications, Mode of communications, LAN, MAN, WAN, Internet, network hardware, network software, Design issues of layers, reference models: OSI, TCP/IP layers, and characteristics.
Physical Layer: Network devices, Network topology, Transmission media, Signal transmission, Network performance and transmission impairments, the public switched telephone network, Switching techniques.

Unit 2	**Data Link Layer**	**9 hours**

Data Link layer: Design issues, Framing, Error Detection and Correction, Flow control (Elementary Data Link Protocols, Sliding Window protocols).
Medium Access Control and Local Area Networks: Multiple access protocols: Random Access (ALOHA, CSMA, CSMA/CD, CSMA/CA), Controlled Access (Polling, Reservation, Token Passing), Channelization (FDMA, TDMA, CDMA). IEEE standards Wired LANs (Ethernet): Traditional Ethernet, Fast Ethernet, Gigabit Ethernet, Wireless LANs: IEEE 802.11 and Bluetooth.

Unit 3	**Network Layer**	**9 hours**

Network Layer: Network layer design issues, Logical addressing: Basic internetworking, Subnetting, Supernetting, NAT, network layer protocols, IPv4, IPv6
Routing: Static and dynamic routing, Routing algorithms and protocols (Shortest Path Algorithms, Flooding, Distance Vector Routing, Link State Routing, Hierarchical Routing, Broadcast Routing, Multicast Routing, Anycast Routing: RIP, OSPF, BGP)

Unit 4	**Transport Layer**	**9 hours**

Transport Layer: Process-to-process delivery, Transport layer protocols (UDP and TCP), SCTP, Connection management,
Congestion control in TCP: Congestion Control: Open Loop, Closed Loop choke packets; Quality of service: techniques to improve QoS: Leaky bucket algorithm, Token bucket algorithm

Unit 5	**Application Layer**	**9 hours**

Session Layer: Roles and responsibility of session layer protocols
Presentation Layer: Cryptography Symmetric-key cryptography, Asymmetric-key cryptography, Digital Signature, Compression: Lossless and Lossy Compression
Application Layer: Domain Name System, World Wide Web and Hyper Text Transfer Protocol, Electronic mail, File Transfer Protocol, Simple Network Management Protocol, Telnet.

Total Lecture Hours	**45 hours**

Textbook:

1. Behrouz Forouzan, “Data Communication and Networking” Fourth Edition-2006, Tata McGraw Hill
2. Andrew Tanenbaum “Computer Networks”, Fifth Edition-2011, Prentice Hall.
3. William Stallings, “Data and Computer Communication”, Eighth Edition-2008, Pearson.

Reference Books:

1. Kurose and Ross, “Computer Networking- A Top-Down Approach”, Eighth Edition-2021, Pearson.
2. Peterson and Davie, “Computer Networks: A Systems Approach”, Fourth Edition-1996, Morgan Kaufmann
3. Behrouz Forouzan, “TCP/IP Protocol Suite”, McGraw Hill.

Mode of Evaluation:

<table>
<tr><th colspan="4">Evaluation Scheme</th></tr>
<tr><th>MSE</th><th>CA</th><th>ESE</th><th>Total Marks</th></tr>
<tr><td>60</td><td>15</td><td rowspan="2">75</td><td rowspan="2">150</td></tr>
<tr><td colspan="2">75</td></tr>
</table>

Course Code: HS112L	**Course Name: Universal Human Values**	**L**	**T**	**P**	**C**
		3	**0**	**0**	**3**

Pre-requisite: NIL

Course Objectives:

1. To help students distinguish between values and skills, and understand the need, basic guidelines, content, and process of value

education.
2. To help students initiate a process of dialog within themselves to know what they really want to be in their life and profession
3. To help students understand the meaning of happiness and prosperity for a human being.
4. To facilitate the students to understand harmony at all the levels of human living, and live accordingly.
5. To facilitate the students in applying the understanding of harmony in existence in their profession and lead an ethical life.

Course Outcome: After completion of the course, the student will be able to
1. Understand the significance of value inputs in a classroom, distinguish between values and skills, understand the need, basic guidelines, content, and process of value education, explore the meaning of happiness and prosperity, and do a correct appraisal of the current scenario in the society.
2. Distinguish between the Self and the Body, and understand the meaning of Harmony in the Self and the Co-existence of Self and Body.
3. Understand the value of harmonious relationships based on trust, respect, and other naturally acceptable feelings in human-human relationships and explore their role in ensuring a harmonious society.
4. Understand the harmony in nature and existence, and workout their mutually fulfilling participation in nature.
5. Distinguish between ethical and unethical practices, and start working out the strategy to actualize a harmonious environment wherever they work.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	2	2	1	-	1	1
CO2	-	-	-	-	-	2	2	1	-	1	1
CO3	-	-	-	-	-	2	2	1	-	1	1
CO4	-	-	-	-	-	2	2	1	-	1	1
CO5	-	-	-	-	-	2	3	1	-	1	1

Unit 1 | Introduction to Value Education | 09 hours

Understanding the need, basic guidelines, content, and process for Value Education, Self- Exploration–what is it? - its content and process; 'Natural Acceptance' and Experiential Validation –as the mechanism for self-exploration, Continuous Happiness, and Prosperity-A look at basic Human Aspirations, Right understanding, Relationship, and Physical Facilities-the basic requirements for fulfillment of aspirations of every human being with their correct priority, Understanding Happiness and Prosperity correctly – A critical appraisal of the current scenario, Method to fulfill the above human aspirations: understanding and living in harmony at various levels.

Unit 2 | Understanding Harmony in Human Being | 09 hours

Understanding human being as a co-existence of the sentient 'I' and the material 'Body', Understanding the needs of Self ('I') and 'Body' - Sukh and Suvidha, Understanding the Body as an instrument of 'I' (I being the doer, seer, and enjoyer), Understanding the characteristics and activities of 'I' and harmony in 'I', Understanding the harmony of I with the Body: Sanyam and Swasthya; correct appraisal of Physical needs, the meaning of Prosperity in detail, Programs to ensure Sanyam and Swasthya.

Unit 3 | Understanding Harmony in the Family and Society | 09 hours

Harmony in Human-Human Relationship Understanding harmony in the Family-the basic unit of human interaction, Understanding values in the human-human relationship; meaning of Nyaya and program for its fulfillment to ensure Ubhay-tripti; Trust (Vishwas) and Respect(Samman) as the foundational values of relationship, Understanding the meaning of Vishwas; Difference between intention and competence, Understanding the meaning of Samman, Difference between respect and differentiation; the other salient values in a relationship, Understanding the harmony in the society (society being an extension of the family): Samadhan, Samridhi, Abhay, Sah- astitva as comprehensive Human Goals, Visualizing a universal harmonious order in society- Undivided Society (Akhand Samaj), Universal Order (Sarvabhaum Vyawastha) – from family to world family!

Unit 4 | Understanding Harmony in Nature and Existence | 09 hours

Whole existence as Co-existence Understanding the harmony in Nature, Inter connectedness, and mutual fulfillment among the four orders of nature- recyclability and self-regulation in nature, Understanding Existence as Co-existence (Sah-astitva) of mutually interacting units in all- pervasive space, Holistic perception of harmony at all levels of existence.

Unit 5 | Implications of the above Holistic Understanding of Harmony on Professional Ethics | 09 hours

Natural acceptance of human values, Definiteness of Ethical Human Conduct, Basis for Humanistic Education, Humanistic Constitution and Humanistic Universal Order, Competence in Professional Ethics.

Total Lecture Hours | 45 hours

Textbook:

1. R R Gaur, R Asthana, G P Bagaria, 2019 (2nd Revised Edition), A Foundation Course in Human Values and Professional Ethics. ISBN 978-93-87034-47-1, Excel Books, New Delhi.

Reference Books:

1. Ivan Illich, Energy & Equity, The Trinity Press, Worcester and Harper Collins, USA,1974.
2. E.F. Schumacher, Small is Beautiful: a study of economics as if people mattered, Blond & Briggs, Britain,1973.
3. A Nagraj, Jeevan Vidya EkParichay, Divya Path Sansthan, Amarkantak 1998.
4. P L Dhar, RR Gaur, Science and Humanism, Commonwealth Publishers 1990.

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
60	15	75	150
75			

Course Code: HS113L	Course Name: Aptitude-2	L	T	P	C
		1	0	0	1

Pre-requisite: Should have cleared Aptitude – 1 offered in the 3rd semester

Course Objectives:

1. To provide adequate exposure to the students regarding the use of aptitude tests in the recruitment process and competitive examinations.
2. To improve the logical & numerical ability of the students.

Course Outcome: After completion of the course, the student will be able to

1. Explain the logical structures and spatial reasoning concepts involved in analytical aptitude problem solving.
2. Apply deductive reasoning techniques to interpret statements and establish logical relationships for decision making.
3. Analyze time-based and sequential problem scenarios to derive accurate solutions using logical and mathematical reasoning.
4. Evaluate quantitative and critical information from multiple data sources to support logical judgments and informed decisions.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	1	-	1	-	1	-	-	-	-	1
CO2	1	1	-	1	-	2	-	-	-	-	1
CO3	1	1	-	1	-	1	-	-	-	-	2
CO4	2	1	-	1	-	1	-	-	-	-	1

Unit 1	Analytical Reasoning & Logical Puzzles	04 hours

Definition and Introduction of Concept and Relation of Cube and Cuboids, Cut the cube in different layer and then solve questions accordingly. Problems related with open and closed dice.

Unit 2	Syllogism	03 hours

Understanding of Venn diagram, Problems related with Venn diagram, Statement and Conclusion, Syllogism and reverse syllogism.

Unit 3	Clock and Calendar	04 hours

Definition and Introduction of Concept and Relation of angle and time, Overtaking, overlapping, right-angle and straight Angle with respect to time, Error in clock (faster and slower), Correct time of clock, Mirror and Water Image of clock, Introduction of Calendar, Concept of Normal and Leap Year, Finding Odd days, Finding the day of the week of given date with and without reference.

Unit 4	Data Interpretation and Critical Reasoning	03 hours

Tables (Understand of Table, Fillers in table), Line Graph (Understand the graph, Percentage change, Ratio based comparison), Bar Graph (Type of Bar Graph, Average and Comparison, Stacked Bar Graph), Pi Chart (Conversion of Percentage and Degree, Fillers in Pie chart, Multiple Pie chart), Mixed Graph (problems related with combination of various charts) Critical Reasoning: Assumptions, Cause and Effect, Assertion and Reason, Statement and Inference

Total Lecture Hours | **15 hours**

Textbook:

1. A Modern Approach to Verbal & Non-Verbal Reasoning" by R.S. Aggarwal, S. Chand Publication.
2. https://www.geeksforgeeks.org/most-important-aptitude-topics-for-placements/

Reference Books:

1. "How to Prepare for Logical Reasoning for the CAT" by Arun Sharma, TMH Publication
2. https://www.indiabix.com/logical-reasoning/questions-and-answers/
3. https://testbook.com/placement-aptitude/test-series

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
-	25	25	50
25			

Course Code: HS114L	Course Name: Soft Skills Essentials-2	L	T	P	C
		1	0	0	NC

Pre-requisite: Successful completion of the subject 'Soft Skills Essentials-1' of the third semester.

Course Objectives:

To strengthen students' professional communication, cultural intelligence, and emotional awareness through advanced speaking activities, scenario-based discussions, and digital literacy tasks, equipping them for diverse workplace interactions

Course Outcome: After completion of the course, the student will be able to

1. Apply advanced communication strategies that include vocabulary enhancement.
2. Use the storytelling tecCO4hnique for displaying effective communication in teams
3. Demonstrate prompt writing for AI-based tools and create effective elevator pitches to convey ideas with clarity and impact.
4. Exhibit interpersonal effectiveness by navigating negotiation, persuasion, and emotional intelligence in professional contexts

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	1	-	1	3	-	2
CO2	-	-	-	-	-	-	-	1	3	-	2
CO3	-	-	-	-	-	1	-	1	3	-	2
CO4	-	-	-	-	-	1	-	1	3	-	2

Unit	Title	Hours
Unit 1	**Advanced Communication and Cultural Sensitivity**	**07 hours**
Vocabulary Enhancement through Gamification, Story Coining and Presentations Understanding Cross-Cultural, Communication (DEI) using Case Studies, Duo Lingo English Proficiency Tests		
Unit 2	**Professional Expression and Digital Literacy**	**04 hours**
TMAY through Driver's test, Writing Effective Prompts on Various LLMs, Duo Lingo English Proficiency Tests		
Unit 3	**Interpersonal Effectiveness and Emotional Intelligence**	**04 hours**
Negotiation & Persuasion Role Plays, Developing Emotional Intelligence via Scenario-Based Discussions		
	Total Lecture Hours	**15 hours**

Textbook:

1. https://youtu.be/5Wr-uaGzY7c
2. https://youtu.be/NcCwlqBapHo
3. https://youtu.be/SKNmQPIBPIg
4. RAISEC - B. Tech. MCA - Introduction
5. RAISEC - B. Tech. MCA - Social Personality Type
6. RAISEC - B. Tech. MCA - Enterprising Personality Type
7. RAISEC - B. Tech. MCA - Conventional Personality Type

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
-	25	**25**	**50**
25			

Course Code: IT202B	Course Name: Data Analytics	L	T	P	C
		2	0	2	3

Pre-requisite: Basic Knowledge of Python

Course Objectives:

1. The objective of this course is to introduce key concepts, lifecycle, and tools in data analytics.
2. The aim is to build practical skills in data preprocessing, visualization, and machine learning techniques.
3. The course also focuses on exploring big data technologies such as Hadoop and Spark.

Course Outcome: After completion of the course, the student will be able to

1. Explain key concepts, tools, and applications of data analytics.
2. Apply data cleaning and preprocessing using Python libraries.
3. Analyze data and visualize insights with BI tools.
4. Implement ML algorithms for classification and clustering.
5. Explore Big data framework and demonstrate Hadoop processing.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	2	1	-	-	-	-	-	-	-	1
CO2	2	3	3	2	2	-	-	-	-	1	2
CO3	2	2	3	2	3	1	-	-	-	1	2
CO4	2	3	3	3	3	1	-	-	-	1	2
CO5	2	3	3	3	2	1	-	2	-	1	2

Unit	Title	Hours
Unit 1	**Introduction to Data Analytics**	**12 hours**
Data and types of data (structured, unstructured, semi-structured), characteristics of data, need of data analytics, data analytic tools, applications of data analytics, Data Analytics Lifecycle (DALC), key roles for successful analytic projects.		
Unit 2	**Data Collection, Cleaning & Preprocessing**	**12 hours**

Sources (Primary and Secondary), handling missing data, outliers, noise, transformation, normalization and data wrangling using libraries like Pandas and NumPy.		
Unit 3	**Exploratory Data analysis & Visualization**	**12 hours**
Statistics and Probability for Data Analysis: Distributions, mean, standard deviation, median, and mode, Statistics that are inferred: Confidence intervals, regression analysis, and hypothesis testing, Ideas in Probability: Random variables, probability distributions, and the Bayes Theorem. Data Visualization Tools: Overview of Power BI & Tableau: Generating dynamic reports and dashboards. Interactive Case Studies: Using visualization approaches to solve practical business issues.		
Unit 4	**Machine learning for Data Analytics**	**12 hours**
Introduction to supervised learning and unsupervised learning, Classification models: Decision Trees, k-NN, Naive Bayes, Clustering algorithms: K-Means, Hierarchical Clustering, Dimensionality reduction: PCA, Association rule mining (Apriori algorithm), Model evaluation: Confusion Matrix, Accuracy, Precision, Recall, F1-Score.		
Unit 5	**Big Data and Advanced Analytics**	**12 hours**
Introduction to Big Data, Characteristics of Big data, Big data challenges, Example of big data in real life, Hadoop, Hadoop components, Hadoop ecosystem, Overview of Apache Spark, Pig, Hive, Hbase and Sqoop, Map reduce paradigm: algorithm MapReduce, Matrix vector multiplication by MapReduce.		
Total Lecture Hours		**60 hours**

Textbook:

1. Michael Berthold, David J. Hand, Intelligent Data Analysis, Springer
2. Anand Rajaraman and Jeffrey David Ullman, Mining of Massive Datasets, Cambridge University Press.
3. John Garrett,Data Analytics for IT Networks : Developing Innovative Use Cases, Pearson Education.

Reference Books:

1. Foster Provost, Tom Fawcett, Data Science for Business, O'Reilly Media.
2. Wes McKinney, Python for Data Analysis, O'Reilly Media.
3. Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, An Introduction to Statistical Learning, Springer.
4. Jiawei Han, Micheline Kamber, Jian Pei, Data Mining: Concepts and Techniques, Morgan Kaufmann.
5. Nathan Marz, James Warren, Big Data: Principles and Best Practices of Scalable Real-Time Data Systems, Manning Publications.
6. Tom White, Hadoop: The Definitive Guide, O'Reilly Media.

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
60	15	**75**	**150**
75			

Course Code: CS303B	**Course Name: ANN and Machine Learning**	**L**	**T**	**P**	**C**
		2	**0**	**2**	**3**

Pre-requisite: Basic Knowledge of Python

Course Objectives:

1. To equip students with foundational and advanced Machine Learning (ML) techniques, including data preprocessing, supervised/unsupervised algorithms, and ensemble methods.
2. To provide hands-on experience in Artificial Neural Networks (ANN) and real-world applications like disease prediction, AutoML, and digit recognition.

Course Outcome: After completion of the course, the student will be able to

1. Preprocess, clean, and transform incomplete/noisy datasets and implement regression models for predictive analysis.
2. Implement and evaluate supervised classification algorithms using appropriate performance metrics and visualization tools.
3. Apply unsupervised clustering, outlier detection, and feature engineering/dimensionality reduction techniques to real-world datasets.
4. Design advanced ensemble models and optimize hyperparameters using traditional, metaheuristic, and AutoML frameworks.
5. Construct and simulate Artificial Neural Network (ANN) architectures like MLP and RBF for complex classification and recognition tasks.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	3	3	1	-	-	-	-	2
CO2	2	3	3	2	3	1	-	-	-	-	2
CO3	2	3	3	2	3	2	-	-	-	-	2
CO4	3	2	3	2	3	1	-	-	-	-	2
CO5	3	2	3	2	3	1	-	-	-	-	2

Unit 1	**Understanding the Basics of ML**	**15 hours**
Date pre-processing and transformation techniques implementation- Definition and importance of data pre processing in Machine Learning. Handling Missing values. Encoding categorical variables. Cleaning and transforming an incomplete and noisy dataset. Supervised ML algorithms- Implementing Classification Algorithms in Python – KNN. Implementing Regression Algorithms in Python. House price prediction using Linear Regression. Spam detection using SVM		

Performance metrics- Confusion matrix. Understanding various performance metrics. Implementing various data visualization techniques Mini Projects: Performance Comparison of Classification models on UCI dataset.		
Unit 2	**Practical Introduction to Machine Learning Algorithms**	**15 hours**
Introduction- Feature engineering. Understanding and Implementing feature selection techniques. Implementing PCA for feature extraction Outlier Detection- Understanding the concept of outliers. Identifying and removing outliers using Isolation Forest Unsupervised ML algorithms- Implementing Clustering Algorithms in Python. Document Clustering using DBSCAN, Optics Mini Projects: Feature Selection in Clustering: A Comparative Implementation and Performance Analysis		
Unit 3	**Ensemble Learning and Advanced Topics**	**15 hours**
Hyperparameter Tuning and Optimization- Traditional vs Metaheuristic optimization techniques for tuning. Tuning the hyperparameters of ML algorithms using Grid search Ensemble Learning Algorithms- Building Ensemble models using different techniques- Decision Tree and Random Forest. Implementing ensemble stacking for sentiment analysis Real World applications for ML- Weather Forecasting using a Regression model. Disease prediction using ensemble techniques AutoML- Popular AutoML frameworks and tools. Implementing various AutoML tools for real-world problems Mini Projects: Evaluating AutoML and Manual Hyperparameter Tuning for Disease Prediction Systems		
Unit 4	**Fundamentals of (ANN)**	**15 hours**
Understanding Neural Network- Basics of ANN: Single perceptron, perceptron learning rule. Introduction to Activation functions. Implementing a binary classifier using a single perceptron Neural Network Algorithms with Implementation- Understanding and Implementing MLP in Python. Understanding and Implementing RBFClassifier in Python. Running various simulations of MLP/RBFclassifier using sci-kit ANN Real-life applications with ML libraries- Predicting student performance using MLP. Classifying Music genres using RBFClassifier Mini Project: Handwritten Digit Recognition using MLP		
Total Lecture Hours		**60 hours**

Textbook:

1. Tom M. Mitchell, ―Machine Learning, McGraw-Hill Education (India) Private Limited, 2013.
2. Ethem Alpaydin, ―Introduction to Machine Learning (Adaptive Computation and Machine Learning), MIT Press 2004.
3. Stephen Marsland, ―Machine Learning: An Algorithmic Perspective, CRC Press, 2009.
4. Bishop, C., Pattern Recognition and Machine Learning. Berlin: Springer-Verlag.
5. M. Gopal, "Applied Machine Learning", McGraw Hill Education

Reference Books:

1. "Python Data Science Handbook" – Jake VanderPlas practical for data preprocessing.
2. "Ensemble Methods for Machine Learning" – Gautam Kunapuli (Manning)
3. "AutoML: Methods, Systems, Challenges" – Frank Hutter et al.
4. "Neural Networks and Deep Learning" – Charu C. Aggarwal (Springer)

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
60	15	**75**	**150**
75			

Course Code: CS208B	Course Name: Web Technology	L	T	P	C
		2	0	2	3

Pre-requisite: Web Development, HTML, CSS, JavaScript

Course Objectives:
1. To familiarize students with the modern technologies used in web technologies.
2. To realize the fundamentals of full stack development & python web development frameworks.

Course Outcome: After completion of the course, the student will be able to
1. To understand advanced ES6 JavaScript features to develop modular and asynchronous applications.
2. To apply Dart and Flutter with OOP and async concepts to build cross-platform mobile applications.
3. To design and implement interactive Flutter applications using state management, navigation, and basic animations.
4. To develop web applications using Flask by implementing dynamic routing, template rendering, form validation, and developing lightweight REST APIs.
5. To develop enterprise web applications using Django by applying MVC architecture, managing database models and migrations, and implementing secure form handling.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	2	-	2	-	-	-	-	-	2
CO2	3	2	3	-	2	-	-	2	-	-	2
CO3	2	2	3	-	2	-	-	2	1	-	2
CO4	3	3	2	2	2	-	-	2	1	-	2
CO5	3	3	2	2	2	-	-	2	1	-	2

Unit 1	MERN STACK: NODE.JS	15 hours
Internet and WWW. Introduction to a web-app, basics of web server, fundamental web governing protocols. Introduction to MERN, Introduction to Node.js, Installation Process, blocking vs non-blocking architecture, npm, Event Loop, REPL, concepts of callbacks, Asynchronous Programming, WebSockets, events and EventEmitter, Node.js Streams, file system, http, url, os, console, error handling.		
Unit 2	**MERN STACK: EXPRESS, REACT, DEPLOYMENT**	**15 hours**
Introduction to Express, features and applications, Postman API platform, Routing (GET, POST, PUT, DELETE), concept of middle wares, Templating Engines (EJS, PUG, Handlebars), serving static files with express, Error Handling in Express, React.js: Components, Hooks, State Management (Redux, Context API), Stateless Components, Designing Components, State vs. Props, Connecting React frontend with Express.js backend.		
Unit 3	**MERN STACK: REST AND MONGO**	**15 hours**
Introduction to RESTful API, principles, HTTP Methods in REST, SQL vs NoSQL databases, Introduction to MongoDB: installation, datatypes, collections, documents, Create Database, Create Collection, installing and using mongoose Insert, delete, update, join, sort, query, CRUD operations in MongoDB. Connect web application to DBMS.		
Unit 4	**WEB DEVELOPMENT WITH PYTHON: DJANGO & FLASK**	**15 hours**
Introduction to Flask: Installation, basic routing, request and response handling, Jinja2 templating, form handling, validation, Building REST APIs using Flask and Flask-RESTful. **Introduction to Django**: MVC architecture, installation, project setup, connecting to databases, Creating models, migrations, querying the databases, Django Views & Templates, URL routing, template rendering, form handling		
Total Lecture Hours		**60 hours**

Textbook:
1. Beginning MERN Stack, Greg Lim.
2. Pro MERN Stack, Vasan Subarmanian
3. Two Scoops of Django, by Daniel Roy Greenfeld, Audrey Roy Greenfeld
4. Flask Web Development: Developing Web Applications with Python, Miguel Grinberg

Reference Books:
1. Ultimate Full-Stack Web Development with MERN: Design, Build, Test and Deploy Production-Grade Web Applications with MongoDB, Express, React and NodeJS, Nabendu Biswas
2. Mastering Flask Web Development: Build enterprise-grade, scalable Python web applications , Second Edition, Gaspar Jack Stouffer

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
60	15	**75**	**150**
75			

Professional Elective-I Courses Detail Syllabus – IV Sem

Course Code: CS318E	Course Name: Frontend Engineering with React & Next	L	T	P	C
		3	0	2	4

Pre-requisite: Computer Fundamentals

Course Objectives:
The objective of this course is to provide learners with a strong foundation in modern frontend engineering using React and Next.js. The course enables students to design, develop, and deploy scalable, component-driven user interfaces, implement state management and routing, integrate APIs and AI-powered services, and apply best practices for performance, security, and deployment of production-grade frontend applications.

Course Outcome: After completion of the course, the student will be able to
1. Understand modern frontend architecture, React fundamentals, and component-based UI development principles.
2. Develop dynamic and interactive user interfaces using JSX, state, props, hooks, and reusable components.
3. Implement scalable state management, routing, and data flow using modern React and Redux patterns.
4. Build server-rendered and statically generated web applications using Next.js with authentication and API integration.
5. Deploy, monitor, and optimize frontend applications following industry best practices for performance and scalability

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	-	2	-	-	-	-	-	2
CO2	3	3	2	2	3	-	-	1	1	-	2
CO3	3	3	3	2	3	-	-	2	1	1	2
CO4	3	3	2	3	3	-	-	2	1	2	2
CO5	2	2	2	2	3	-	-	2	2	2	3

Unit 1	Foundations of React ES6	9 hours
Features of React, Practical Application, Why need React, React architecture and mental model, How React Works, Leveraging Virtual DOM, Setting up React, Array & object methods		

Unit 2	JSX	9 hours
Why JSX, Embedding JavaScript, Expression in JSX, JSX as an Expression, Nested elements in JSX, JSX Attributes, JSX Comments, JSX Styling and representation as object, The State of the Component, Defining State, Changing the State, Props, Validation, Validators		
Unit 3	**Elements**	**9 hours**
Rendering Element, About render (), Creating React Element, Updating Element, components, Introducing Components, Types of Components, Functional Component, Functional Components as Stateless, Using Functional Component		
Unit 4	**Redux**	**9 hours**
Component design patterns, Lifting state and prop drilling, Context API for global state, Redux fundamentals: Core principles, Store, actions, reducers, Data flow, Redux Toolkit (modern Redux), Middleware and async logic (Thunk), Debugging with Redux DevTools, Integrating Redux with React, Testing React components basics		
Unit 5	**NextJS**	**9 hours**
Next.js fundamentals: pages, routing, and data fetching (SSR/SSG), Authentication (JWT, OAuth) and session management, Integrating AI/LLM APIs for intelligent UI features, Analytics, logging, and observability for frontend apps, Deployment using Vercel, Netlify, or AWS Amplify		

Lab Practical's **30 hours**

Lab 1: Write JavaScript programs to perform conditional logic, loops, and array manipulations (map, filter, reduce).
Lab 2: Create functions to handle user input, validate data, and display results dynamically using DOM manipulation.
Lab 3: Demonstrate event handling by building a mini to-do list that adds, edits, and deletes tasks.
Lab 4: Implement ES6 features (arrow functions, destructuring, classes, spread/rest) in refactoring an existing JavaScript project.
Lab 5: Set up a React environment using Vite or Create React App and render your first component on the screen.
Lab 6: Build a component-based calculator app using React state and props to manage user interactions.
Lab 7: Create a form in React that uses controlled components, state hooks, and validation using simple regex or custom validators.
Lab 8: Develop a React component hierarchy for a movie or product listing app, passing data through props and maintaining local state.
Lab 9: Integrate Redux in a shopping cart app — manage actions, reducers, and a global store with the Redux DevTools. Lab 10: Combine React and Redux Toolkit to handle asynchronous API calls for fetching product or user data.
Lab 11: Create a single-page application using React Router for navigation between pages (Home, About, Contact).
Lab 12: Implement authentication flow using JWT in a Next.js app (login, logout, session persistence).
Lab 13: Integrate an AI API (like OpenAI or Hugging Face) in a Next.js app for text generation or sentiment analysis.
Lab 14: Deploy your final Next.js project (SSR/SSG) to Vercel or Netlify with environment variables and analytics tracking enabled.

Total Lecture Hours **75 hours**

Textbook:
1. Banks, A., & Porcello, E. Learning React. O'Reilly Media.
2. Grider, S. Modern React with Redux. Udemy Press.

Reference Books:
1. Freeman, E. Full Stack React. O'Reilly Media.
2. Vercel Documentation. Next.js Official Docs.

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
80	20	100	200
100			

Course Code: CS307E	Course Name: Intelligent Systems with Text & Vision API	L	T	P	C
		3	0	2	4

Pre-requisite: Python Programming, Linear Algebra Basics, Probability Fundamentals

Course Objectives:
The objective of this course is to provide learners with a strong foundation in building intelligent systems using Natural Language Processing and Computer Vision techniques. The course enables students to design, implement, and integrate text and vision-based AI models using modern APIs and deep learning frameworks, while developing practical skills for real-world AI-driven applications.

Course Outcome: After completion of the course, the student will be able to
1. Understand the fundamentals of Natural Language Processing and Computer Vision used in intelligent systems.
2. Apply word embeddings, sequence models, and transformer-based approaches for text understanding tasks.
3. Implement computer vision pipelines for image processing, object detection, and visual analysis.
4. Develop multimodal AI applications by integrating text and vision models using APIs and pre-trained frameworks.
5. Evaluate, deploy, and optimize intelligent systems while considering performance, ethical, and real-world constraints

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	-	2	-	-	-	-	-	2
CO2	3	3	2	2	3	-	-	1	1	-	2
CO3	3	3	3	2	3	-	-	2	1	1	2
CO4	3	3	3	3	3	-	-	2	1	2	2
CO5	2	2	2	2	3	-	-	2	2	2	3

Unit 1	Natural Language Processing	9 hours
Introduction to Natural Language Processing, Types of NLP systems, How computer understands text, Terminologies used in NLP, Steps Involved in NLP, Steps involved in pre- processing, Pipeline of NLP Problems o Challenges in NLP **Purpose of DevOps** Purpose of DevOps, Minimum Viable Product (MVP), Benefits of MVP, Application Deployment, Automated Application Deployment, Application Release Automation (ARA), Components of Application Release Automation (ARA), Continuous Integration, Best Practices of CI, Benefits of CI, Continuous Delivery. *Case Study: Building a basic sentiment analysis system for customer feedback data.*		
Unit 2	Words & Vectors	9 hours
Concepts of words and vectors, Techniques of converting words to numbers, GloVe Word Embeddings, Word2Vec and its types, such as Skip Gram, Model and Continuous BOW o Advanced word vectors, limitations of CBOW and Skip Gram, Word window classification, Dependency parsing, Constituency parsing o Machine translation, Attention, End to end models for speech processing, Deep learning for speech recognition, Tree recursive neural networks o RNN for language modelling, Dynamic neural network for question answering. *Case Study: Developing a keyword extraction and semantic similarity engine using word embeddings*		
Unit 3	Introduction To Computer Vision And Image Processing	9 hours
Image Processing, Elements of Image Processing System, Computer Vision, Computer, Graphics, Application Areas, Imaging Geometry, Image Sampling, Mathematical Tools, Image transformations: 2D and 3 D Transformation, Image Enhancements-Intro, Image Segmentation Intro, Cognitive Aspects of Color, VR/AR, Object Recognition, Object Tracking *Case Study: Designing an object detection pipeline for image-based classification tasks*		
Unit 4	Introduction To Python Open CV	9 hours
Introduction, GUI Features, Operations: Pixel Editing, Geometric Transformations, Feature Detection, Video Analysis and Tracking, Stereo Imaging, Calibration, OpenCV-Python, Visualizations, Image Denoising, Object Detection, Transformation and Spatial Filtering Introduction, Functions, Histogram, Histogram Equalization, Histogram Matching (Specification), Local Histogram Processing, Using Histogram Statistics for Image Enhancement Introduction to Spatial Filtering, Smoothing & Sharpening Image Filters *Case Study: Building a multimodal AI application for image captioning and visual question answering.*		
Unit 5	Object Recognition & Tracking	9 hours
Shape correspondence and shape matching, Sliding Window Method, Patterns, Structural Methods, Deformable Objects, Tracking, Strategies, Matching, Tracking with Filters, Data Association, Particle Filtering, Regularization theory, Optical computation, Stereo Vision, Motion estimation, Structure from motion		
Lab Practical's		**30 hours**
Lab 1: Load and preprocess a multimodal dataset containing image captions. Perform text cleaning (tokenization, stopword removal, lemmatization) using spaCy or NLTK, and apply basic image transformations (resize, grayscale) using OpenCV. **Lab 2:** Extract text from images using Tesseract or EasyOCR and perform text preprocessing, punctuation removal, and spell correction on the extracted text. **Lab 3:** Generate keywords or named entities from image captions using NLTK or spaCy and visualize them as word clouds to understand dominant textual themes. **Lab 4:** Perform sentiment analysis on product reviews containing both text and images using a pre trained transformer model (BERT) and image classifier (ResNet/VGG16). Compare correlations between visual aesthetics and sentiment polarity. **Lab 5:** Build a text-based image retrieval system using the CLIP (Contrastive Language–Image Pretraining) model. Given a text prompt, retrieve the most semantically similar images from a dataset. **Lab 6:** Implement a Visual Question Answering (VQA) system using a Vision-Language API such as Hugging Face's VQA model or LLaVA to answer natural language questions about an image. **Lab 7:** Generate image captions using a CNN–LSTM model or pre-trained image captioning API. Evaluate caption quality using BLEU or ROUGE scores. **Lab 8:** Detect and label objects in images using YOLOv8 or Detectron2, and enrich each detected object's label with descriptive context using Word2Vec similarity or a GPT API. **Lab 9:** Combine facial emotion recognition (OpenCV + DeepFace) with text sentiment analysis (TextBlob or BERT) to perform multimodal emotion classification from video interviews. **Lab 10:** Detect objects and scene changes in a short video and automatically generate a natural language summary using a pre-trained summarization model such as T5 or BART. **Lab 11:** Capture handwritten notes using a webcam, convert them to text using OCR, and perform NLP-based topic classification or summarization using transformer APIs. **Lab 12:** Implement a multimodal fake news detector by combining textual headline analysis (DistilBERT) and visual content cues (CNN features). Integrate predictions to classify posts as real or fake. **Lab 13:** Detect text from images (e.g., signboards) using OCR and translate it to another language using Google Translate API or MarianMT, then overlay the translated text on the original image. **Lab 14:** Build an interactive multimodal chatbot capable of understanding both image and text inputs. Use Vision-Language APIs (e.g., CLIP or BLIP) to describe, interpret, and answer user queries about uploaded images.		
	Total Lecture Hours	**75 hours**
Textbook: 1. Introduction to Natural Language Processing. Jacob Eisenstein, MIT Press 2. Mastering OpenCV with Python, O'Reilly Media		
Reference Books: 1. Learning Image Processing with OpenCV, Packt Publishing. 2. OpenCV Documentation. OpenCV Official Docs. 3. NLTK Documentation. NLTK Official Docs		

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
80	20	100	200
100			

Course Code: CS304E	Course Name: DevOps Foundations & Version Control	L	T	P	C
		3	0	2	4

Pre-requisite: Programming Fundamentals, Operating Systems Basics, Computer Networks Basics

Course Objectives:
The objective of this course is to provide learners with a strong foundation in DevOps principles, collaborative software development practices, and version control systems. The course enables students to understand the evolution of DevOps, apply Linux and Git-based workflows, implement AI-assisted development practices, and adopt secure, scalable, and automated software delivery processes aligned with modern industry standards.

Course Outcome: After completion of the course, the student will be able to

1. Understand DevOps concepts, agile practices, and the evolution of modern software delivery models.
2. Apply Linux fundamentals and system administration skills for DevOps environments.
3. Implement distributed version control workflows using Git for collaborative software development.
4. Use AI-assisted tools and GitOps practices to enhance code quality, automation, and deployment reliability.
5. Apply security, compliance, and productivity analytics practices in DevOps pipelines

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	-	2	-	-	1	1	-	2
CO2	3	3	2	2	3	-	-	1	-	-	2
CO3	3	3	3	2	3	-	-	2	1	1	2
CO4	3	3	3	3	3	-	-	2	2	2	2
CO5	2	2	2	2	3	1	2	2	2	1	3

Unit 1	Traditional SDLC Models, their problems, & their solutions overview	9 hours

Software, History of Software Engineering and Software, Development Methodologies, Traditional Software Development Models, Waterfall Model, Classical Waterfall Model, Traditional IT Organizations, Developers vs IT Operations Conflict, Birth of Agile, Four Values of the Agile Manifesto, Agile and Lean

Unit 2	Linux Basics and Admin	9 hours

Introduction to Linux (Operating System), Importance of Linux in DevOps, Linux Basic Command Utilities, Linux Administration, Environment Variables, Networking, Linux Server Installation, RPM and YUM Installation

Unit 3	Source Code Management & Git Workflows	9 hours

Centralized vs. distributed SCM systems (SVN vs. Git), Git internals: commits, branches, merges, rebases, Branching strategies: GitFlow, feature branching, trunk-based dev, Pull requests, code reviews, and merge approvals, Create repositories, manage PRs, resolve merge conflicts

Unit 4	AI-Assisted Coding & Intelligent SCM	9 hours

GitHub Copilot / CodeWhisperer basics, AI-based code completion, documentation, and refactoring, Integrating SonarQube + AI for code quality metrics, GitOps introduction: managing infra via Git (FluxCD, ArgoCD), Build a GitOps pipeline to deploy a sample app

Unit 5	Security, Compliance & Productivity Analytics	9 hours

Policy-as-Code using OPA, Snyk AI for pre-commit scans, Secrets management & secure Git workflows, DevEx analytics – tracking productivity and automation ROI, Building dashboards for code quality and release cadence, Implement secure CI trigger using OPA + Jenkins + GitHub

Lab Practical's **30 hours**

Lab 1: Automate Linux server setup using shell scripts to install essential DevOps tools like Git, Docker, and Python.
Lab 2: Create a system monitoring script that checks CPU, memory, and disk usage, and triggers alerts for threshold breaches.
Lab 3: Simulate a multi-developer workflow using Git branches, pull requests, and merge conflict resolution on GitHub.
Lab 4: Configure Git pre-commit hooks to run automated code linting and enforce quality before code submission.
Lab 5: Integrate Git with Jenkins to automatically trigger build pipelines on each code commit.
Lab 6: Implement GitOps deployment using ArgoCD to manage application environments via version control.
Lab 7: Use GitHub Copilot or CodeWhisperer to generate and refactor code in a collaborative development project.
Lab 8: Perform static code analysis with SonarQube and integrate AI-driven suggestions for improving code quality.
Lab 9: Apply Policy-as-Code using Open Policy Agent (OPA) to validate configuration files and enforce repository rules.
Lab 10: Secure your SCM workflow by implementing secret management and vulnerability scanning using Snyk AI.
Lab 11: Version Terraform infrastructure code within Git and collaborate using feature branching and PR workflows.
Lab 12: Visualize team productivity metrics such as commit frequency and review times using Grafana dashboards.
Lab 13: Design a rollback mechanism using Git hooks to automatically revert faulty commits on build failure.
Lab 14: Build a complete GitOps pipeline from code commit to deployment with integrated quality, policy, and security checks.

Total Lecture Hours **75 hours**

Textbook:

1. The DevOps Handbook. O'Reilly Media.
2. The Linux DevOps Handbook. O'Reilly Media

Reference Books:
1. The Phoenix Project
2. Git-SCM Documentation.

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
80	20	100	200
100			

Course Code: CS335E	Course Name: AWS Foundations & Data Engineering	L	T	P	C
		3	0	2	4

Pre-requisite: NA

Course Objectives:
The objective of this course is to provide a strong foundation in cloud computing and data engineering using Amazon Web Services (AWS) by enabling learners to design, deploy, and manage cloud infrastructure, build scalable data pipelines, and apply security and governance best practices, while preparing them for the **AWS Cloud Practitioner – Foundation** and **AWS Data Engineer Associate** certifications.

Course Outcome: After completion of the course, the student will be able to
1. Understand cloud computing concepts, AWS global infrastructure, and the role of cloud platforms in application and data architectures.
2. Configure and manage AWS compute, storage, networking, and identity services to support secure and scalable workloads.
3. Design and implement data ingestion, transformation, and storage pipelines that convert raw data into analytics-ready datasets.
4. Monitor, optimize, and troubleshoot cloud-based data systems to ensure performance, reliability, and cost efficiency.
5. Apply security, governance, and compliance controls to protect data and manage access across cloud-based data platforms.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	-	2	-	-	-	-	-	2
CO2	3	3	2	2	3	-	-	-	-	-	2
CO3	3	3	3	2	3	-	-	-	-	-	2
CO4	3	3	2	3	3	-	-	-	-	-	2
CO5	3	2	2	2	3	1	2	-	-	-	2

Unit 1 | Integrated AWS Cloud Foundations: Compute, Storage, Security, and Deployment | 15 hours

A Comprehensive and integrated understanding of **AWS Cloud Foundations**, covering cloud computing fundamentals, core AWS services, infrastructure design, security best practices, cost management, cloud adoption strategies, and infrastructure as code (IaC).

Case Study: Design and Deployment of a Secure, Scalable Cloud-Based Web Application using AWS

Unit 2 | Cloud Data Ingestion and Pipeline Design | 15 hours

Data ingestion architectures are defined by evaluating source system characteristics, data velocity, reliability requirements, and delivery guarantees. Batch, streaming, and event-driven pipelines are used to move data from operational systems into cloud environments. Data landing zones, message handling, validation, and fault-tolerant delivery mechanisms support continuous and consistent data capture. Emphasis is placed on scalable pipeline design that enables efficient downstream processing and analytics.

Case Study: Building a Real-Time Clickstream Ingestion System using AWS

Unit 3 | Data Transformation, Modeling, and Analytics Preparation | 15 hours

Data transformation workflows are designed to convert raw inputs into structured, consistent, and analytics-ready datasets. Schema management, data cleansing, enrichment, normalization, and aggregation techniques support the preparation of high-quality data. Modeling approaches for analytical workloads enable efficient querying and reporting across large datasets. Emphasis is placed on ensuring data accuracy, consistency, and usability for business intelligence and advanced analytics.

Case Study: Transforming Raw Sales Data into an Analytics-Ready Data Warehouse

Unit 4 | Scalable Data Storage and Operational Management | 15 hours

Data storage and operational strategies are determined by workload characteristics, growth patterns, performance requirements, and cost constraints. Structured, semi-structured, and unstructured data are organized using layered architectures, partitioning methods, and lifecycle policies. Monitoring, throughput optimization, and failure detection maintain pipeline reliability and efficiency. Emphasis is placed on supporting scalable analytics while maintaining durability, availability, and operational stability.

Case Study: Designing and Operating a Multi-Layer Data Lake on AWS

Unit 5 | Data Security, Governance, and Compliance Management | 15 hours

Data protection and governance frameworks are established through identity-based access controls, encryption strategies, auditing mechanisms, and classification policies. Regulatory and organizational compliance requirements guide data handling, retention, and access management. Governance models ensure accountability, traceability, and controlled usage across data pipelines and storage layers. Emphasis is placed on maintaining confidentiality, integrity, and compliance throughout the data lifecycle.

Case Study: Securing and Governing Sensitive Customer Data on AWS

Total Lecture Hours | 75 hours

Textbook:
1. Dua, S. (2021). *AWS Certified Data Analytics Study Guide.* Wiley.
2. Singh, G. (2023). *Data Engineering with AWS.*Packt Publishing

Reference Books:

1. Hellerstein, J., Stonebraker, M., & Hamilton, J. (2019). *Readings in Database Systems (5th ed.).* MIT Press.
2. NIST Big Data Working Group. (2015). *NIST Big Data Interoperability Framework.* NIST Special Publication.

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
80	20	**100**	**200**
100			

Course Code: CS321E	**Course Name: Foundations of iOS App Development**	**L**	**T**	**P**	**C**
		3	**0**	**2**	**4**

Pre-requisite: Basic C++, Object oriented programming

Course Objectives:

1. To familiarize students with the Fundamentals of Swift Programming and iOS Environment Navigation
2. To Design and Build Functional iOS User Interfaces and Applications.

Course Outcome: After completion of the course, the student will be able to

1. To Identify key areas of the Xcode IDE user interface and basic components of an iOS project structure.
2. To Implement core Swift syntax, control flow structures, and error-handling techniques to write functional code.
3. To Apply various Swift data structures (collections, classes, and structs) to select the optimal model for managing data.
4. To Use basic SwiftUI views and UIKit layout constraints to position UI components on a screen.
5. To Execute testing and debugging procedures on an iOS simulator to run a basic application.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	1	1	-	3	-	-	-	-	-	2
CO2	3	3	2	2	2	-	-	-	-	-	2
CO3	3	3	3	2	2	-	-	-	-	-	2
CO4	2	2	3	3	3	-	-	-	-	-	2
CO5	2	3	2	2	3	-	-	-	-	-	2

Unit 1 | Introduction to iOS & Development Environment | 15 hours

Introduction to the iOS Ecosystem: Overview of Apple's ecosystem, developer program, and app lifecycle basics. Xcode Interface: Navigating the IDE, Toolbar, Navigator area, Editor area, Utility area, and Debug area. Project Structure: Understanding AppNameApp.swift, ContentView.swift, Assets catalog, and Info.plist. Preview System: Utilizing Xcode Previews for real-time UI rendering and debugging.

Unit 2 | Swift Fundamentals & Control Flow | 15 hours

Swift Basics: Syntax rules, comments, and printing to the console. Variables & Data Types: Working with Type Inference, Constants (let), Variables (var), Strings, Integers, Floats, Doubles, and Booleans. Control Flow: Conditional statements (if, else if, else), switch statements, and loops (for-in, while). Functions & Closures (Basics): Defining and calling functions, parameters, return types, and an introduction to basic closure syntax. Error Handling: Using try, catch, and throw to manage runtime errors.

Unit 3 | Advanced Swift: Collections & Data Structures | 15 hours

Collections in Swift: Arrays: Ordered lists, indexing, and common array methods. Dictionaries: Key-value pairs, accessing and modifying data. Sets: Unordered collections of unique values. Object-Oriented Swift: Classes vs. Structs: Understanding reference types vs. value types. Enums: Defining enumerations, raw values, and associated values. Advanced Concepts: Optionals: Unwrapping optionals safely using if let and guard let, nil-coalescing. Protocols & Extensions: Defining blueprints with protocols and adding functionality with extensions.

Unit 4 | UI Development (UIKit Essentials & Intro to SwiftUI) | 15 hours

Introduction to SwiftUI: Declarative UI concepts, basic views (Text, Image, Button), and working with colors. UIKit Essentials: Understanding Views (UIView) and View Controllers (UIViewController). Layout & Layout Views: Basic layout constraints, arrangement of views, and alignment. Advanced Views: Working with static and dynamic images. Introduction to TableViews for displaying scrollable lists of data.

Unit 5 | Hands-on Integration & Mini App Project | 15 hours

Project Selection: Choosing a focus track (building a Calculator, a Notes app, or an Image Gallery). App Architecture: Planning the data flow, designing the user interface, and structuring the code. Development and Integration: Combining Swift logic with UI components to build the app features. Testing and Refinement: Debugging, running the app on a simulator, and performing final UI adjustments.

Total Lecture Hours | 75 hours

Textbook:

1. Swift Programming: The Big Nerd Ranch Guide by Mikey Ward
2. SwiftUI Essentials - iOS Edition by Neil Smyth.

Reference Books:

1. iOS Programming Fundamentals with Swift by Matt Neuburg (O'Reilly Media).
2. Thinking in SwiftUI by Chris Eidhof and Florian Kugler (objc.io).

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
80	20	100	200
100			

Course Code: IT306E	Course Name: Microsoft Azure Fundamentals	L	T	P	C
		3	0	2	4

Pre-requisite: NA

Course Objectives:
1. To introduce students to cloud computing concepts with emphasis on Microsoft Azure
2. To enable learners to understand Azure services relevant to data engineering
3. To prepare students for Microsoft Azure Fundamentals (AZ-900) certification
4. To develop industry-ready skills for cloud-based data solutions

Course Outcome: After completion of the course, the student will be able to
1. Explain cloud computing concepts, service models, and Azure architecture
2. Identify and use core Azure services including compute, storage, and networking
3. Apply Azure data services for data storage, processing, and analytics
4. Understand Azure security, identity, governance, and compliance features
5. Evaluate Azure pricing models, SLAs, and manage cloud resources efficiently

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	–	–	2	–	–	–	–	–	2
CO2	3	2	2	–	3	–	–	–	–	–	2
CO3	3	3	3	2	3	–	–	–	–	–	2
CO4	2	2	–	–	2	2	3	–	–	–	2
CO5	2	2	–	–	2	–	3	–	–	3	3

Unit 1 — Introduction to Cloud Computing and Azure — 15 hours
- Evolution of Cloud Computing
- Cloud Service Models: IaaS, PaaS, SaaS
- Cloud Deployment Models: Public, Private, Hybrid
- Overview of Microsoft Azure
- Azure Global Infrastructure (Regions, Availability Zones)
- Azure Portal, Azure CLI, Azure Resource Manager (ARM)

Unit 2 — Core Azure Services – Compute, Storage & Networking — 15 hours
- Azure Compute Services
  - Virtual Machines
  - App Services
  - Azure Functions
- Azure Storage Services
  - Blob Storage
  - File Storage
  - Queue and Table Storage
- Azure Networking Basics
  - Virtual Networks
  - Load Balancer
  - VPN Gateway

Unit 3 — Azure Data Services for Data Engineering — 15 hours
- Azure SQL Database
- Azure Cosmos DB
- Azure Data Factory
- Azure Synapse Analytics (Overview)
- Azure Data Lake Storage
- Basic Data Engineering use cases on Azure

Unit 4 — Security, Identity, and Governance in Azure — 15 hours
- Azure Security Fundamentals
- Azure Active Directory
- Role-Based Access Control (RBAC)
- Azure Policies and Blueprints
- Azure Monitor and Log Analytics

- Compliance and Trust in Azure

Unit 5	Azure Pricing, SLA, and Cloud Management	15 hours

- Azure Pricing Models
- Cost Management and Billing
- Azure Service Level Agreements (SLAs)
- Azure Support Plans
- Resource Optimization and Best Practices
- Overview of AZ-900 Certification Exam Pattern

Total Lecture Hours	**75 hours**

Textbook & Reference Books:
1. Exam Ref AZ-900: Microsoft Azure Fundamentals (Jim Cheshire): The official Microsoft Press guide
2. Learning Microsoft Azure: Cloud Computing and Development Fundamentals (Andersson, Carrio) Oreilly

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
80	20	100	200
100			

Practical Courses Detail Syllabus – IV Sem

Course Code: CS401P	Course Name: Design & Analysis of Algorithms Lab	L	T	P	C
		0	0	2	1

Pre-requisite: The course requires a background in **mathematics** and **strong programming skills.**

Course Objectives:
1. Implement and analyse Greedy, DP, Backtracking, and Branch & Bound techniques for efficient problem-solving.
2. Evaluate the performance and complexity of sorting, graph, and optimization algorithms through hands-on programming.

Course Outcome: After completion of the course, the student will be able to
1. Implement and analyse Greedy, DP, Backtracking, and Branch & Bound for efficient problem-solving.
2. Evaluate algorithm performance and complexity through hands-on programming in sorting, graphs, and optimization.
3. Develop problem-solving skills by implementing Knapsack, Matrix Chain Multiplication, Graph Colouring, and TSP.
4. Understand NP-Completeness and Approximation Algorithms for tackling computationally hard problems.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	3	2	-	-	-	-	-	2
CO2	3	3	3	3	3	-	-	-	-	-	2
CO3	3	3	3	3	3	-	-	-	-	-	2
CO4	3	3	3	3	2	-	-	-	-	-	2

List Of Practical's (Indicative & Not Limited To):
1. Implement and analyze **Merge Sort** using recurrence relations.
2. Implement **Quick Sort** and **Randomized Quick Sort** and compare efficiency.
3. Implement **Shell Sort** and analyze its time complexity.
4. Implement **Counting Sort, Radix Sort, and Bucket Sort**, and compare their performance.
5. Solve the Fractional Knapsack Problem using the Greedy approach.
6. Implement the Activity Selection Problem using Greedy Strategy.
7. Solve Task Scheduling with Deadline and Penalty using Greedy Strategy.
8. Implement Kruskal's Algorithm to find the Minimum Spanning Tree (MST).
9. Implement Prim's Algorithm to find the MST.
10. Implement 0-1 Knapsack Problem using Dynamic Programming.
11. Solve the Optimal Binary Search Tree (OBST) problem.
12. Implement Matrix Chain Multiplication using Dynamic Programming.
13. Solve the Longest Common Subsequence (LCS) Problem.
14. Implement the All-Pairs Shortest Path Algorithm (Floyd-Warshall).
15. Given a set of coin denominations and a target amount **N**, determine the **minimum number of coins** needed and the **total number of ways** to make **N** using the given denominations.
16. Implement the N-Queens Problem using Backtracking.
17. Solve the Subset Sum Problem using Backtracking.
18. Implement Graph Coloring using Backtracking.
19. Solve the Hamiltonian Cycle Problem using Backtracking.
20. Given an **N×N** chessboard, the Knight's Tour problem requires finding a sequence of moves where a knight visits **every square exactly once** without repetition. The knight moves in an **L-shape** (two squares in one direction and one perpendicular).

21. Implement Traveling Salesman Problem (TSP) using Branch and Bound.
22. Implement the Naïve String Matching Algorithm.
23. Implement Rabin-Karp String Matching Algorithm.
24. Implement Knuth-Morris-Pratt (KMP) String Matching Algorithm.
25. Solve the Vertex Cover Problem using Approximation Algorithms.

Total Lecture Hours: 30 hrs.

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
-	25	25	50
25			

Course Code: IT302P	Course Name: Computer Networks Lab	L	T	P	C
		0	0	2	1

Pre-requisite: Computer Fundamentals and C programming

Course Objectives:
The objective of this course is to provide students with a theoretical and practical base in computer networks issues and prepare the students for easy transfer from academia into practical life.

Course Outcome: After completion of the course, the student will be able to
1. Understand the fundamental concepts of computer networking and Network topologies.
2. Analyze different types of network devices and simple computer networks.
3. Implement the basic network commands and use techniques, skills, and modern networking tools necessary for engineering practice.
4. Analyze the impact of computer network technologies on society and professional development.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	2	-	-	2	-	-	-	-	1	2
CO2	3	2	-	-	2	-	-	-	-	1	2
CO3	3	3	2	2	2	-	-	-	2	1	2
CO4	-	2	-	-	-	3	-	-	-	-	3

List Of Practical's (Indicative & Not Limited To):
1. To study of LAN using different Network topologies and different network connecting devices like Hub, Switch, Router etc.
2. To study and learn the handling of networking hardware like: NIC card, RJ-45 connector, crimping tool, CAT-6 cable and Do the following Cabling works in a network (T568A and T568B) a) Cable Crimping b) Standard Cabling and c) Cross Cabling d) IO connector crimping e) Testing the crimped cable using a cable tester.
3. Running and using services/commands like ifconfig, ping, traceroute, nslookup, telnet, ftp, etc.
4. Establish a physical peer to peer network connection using two or more systems using network interface card, network cables, Switch and Router in a LAN.
5. To install the network packet analysis tool Wireshark and also capture and analyze the local traffic.
6. Simulate and Configure a Network topology using packet tracer software.
7. Implement the data link layer framing methods such as character count, character stuffing and bit stuffing.
8. Implement the error correcting code Cyclic Redundancy Check (CRC) of data link layer using various polynomials like CRC-CRC 12, CRC 16 and CRC CCIPP.
9. Implement a program to determine if the IP address is in Class A, B, C, D, or E and translate dotted decimal IP address into 32-bit address.
10. Implement Dijkstra algorithm to compute the shortest path through a given graph.

Total Hours	**30 hours**

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
-	25	25	50
25			